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Multi-scale 3D object semantic segmentation based on Deep Learning

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Abstract

Recent advances in deep learning have significantly improved 3D point cloud semantic segmentation, yet raw point clouds remain sparse, unordered, and highly non-uniform in density. At the same time, common benchmarks exhibit clear semantic class imbalance: large structural categories occupy many more points than small object categories. This project studies a less explored consequence of this imbalance: point sampling is not a neutral preprocessing step, because geometry-driven sampling can discard minority-class evidence before the segmentation backbone is applied. To examine this problem, we establish a unified Point Transformer V3 based evaluation pipeline on ScanNet, S3DIS, and PartNet, and analyse the relationship between class frequency and class-wise segmentation quality. The results show a consistent positive association between point frequency and IoU across all three datasets, indicating that under-represented classes are systematically harder to preserve and segment. Motivated by this observation, we investigate semantic-aware sampling through two complementary settings: an oracle sampler that uses ground-truth labels to reallocate the sampling budget as an upper-bound reference, and a learnable Sampler that predicts point-wise sampling probabilities without access to test labels. Experiments show that semantic-aware sampling improves the quality of the selected sparse subset and benefits several minority or small-object classes. On ScanNet, the learned Sampler approaches the oracle result at the subset level and outperforms grid sampling after downstream retraining. However, the gain on the sparse subset does not transfer perfectly to the full point cloud: KNN aggregation, stochastic multi-mask aggregation, and architecture ablations show that mapping sparse predictions back to dense scenes is a major bottleneck. Overall, the study demonstrates that sampling can be used as an effective intervention point for class imbalance, while also showing that stronger sparse-to-dense aggregation is required to fully convert subset-level gains into full-scene segmentation performance.

Keywords

3D Point Cloud, Semantic Segmentation, Sampling, Transformer

摘要

近年来，深度学习显著推动了三维点云语义分割的发展，但真实点云数据通常具有稀疏、无序和密度不均的特点，同时在语义类别上呈现明显的不平衡分布。本文关注一个常被忽视的问题：点云采样并非中性的预处理步骤，几何驱动的采样策略可能在分割网络处理之前进一步削弱低频类别的几何证据。为研究这一问题，本文基于 Point Transformer V3 在 ScanNet、S3DIS 和 PartNet 上建立统一实验框架，并分析类别频率与逐类分割性能之间的关系。实验表明，三个数据集均存在类别频率与 IoU 的正相关，说明低频类别在采样和分割过程中更容易处于劣势。基于这一观察，本文从两个层面研究语义感知采样：一方面构造使用真实标签重新分配采样预算的 Oracle 采样器，用作语义重分配的上界参考；另一方面训练可学习 Sampler，在测试阶段不使用真实标签的条件下预测点级采样概率。实验结果表明，语义感知采样能够改善所选稀疏子集的类别均衡性，并提升多个低频或小物体类别的分割表现。在 ScanNet 上，可学习 Sampler 在子集层面接近 Oracle 采样器，并在下游重训练后优于传统网格采样。然而，子集性能并不会完全转化为完整点云性能。进一步的 KNN 映射、多次随机掩码聚合和架构消融表明，从稀疏子集预测恢复到完整场景是限制最终性能的重要瓶颈。总体而言，本文表明采样阶段可以作为缓解三维语义分割类别不平衡的有效切入点，同时也说明未来需要更强的稀疏到稠密聚合方法，才能更充分地发挥子集层面的优势。

关键词

3D 点云, 语义分割, 采样算法, Transformer

Chapter 1: Introduction

3D understanding has attracted growing attention in recent years, driven by the increasing availability of depth sensors such as LiDAR and RGB-D cameras [1, 2] and the expanding demands of downstream perception tasks. Compared with 2D images, 3D data preserves complete geometric and spatial information, enabling more precise environment perception [3]. Among various 3D representations—including meshes, voxel grids, and point clouds—point clouds have been widely adopted due to their lightweight structure, flexibility, and ease of acquisition from commodity sensors [4].

3D point cloud semantic segmentation assigns a semantic label to each point in a point cloud, enabling fine-grained understanding of both complex scenes and individual object parts [4, 5]. This capability underpins a range of applications including autonomous driving [6, 7], robotic manipulation [8, 9], and augmented reality [1, 10], where systems must jointly reason about geometry and semantics. Driven by advances in network design, pre-training strategies, and large-scale benchmarks, segmentation accuracy on standard datasets has improved steadily in recent years [5, 11].

Despite steady progress, 3D point cloud semantic segmentation faces challenges that go beyond those encountered in 2D image segmentation. Like its 2D counterpart, point cloud segmentation suffers from semantic class imbalance: benchmarks such as S3DIS [12] and ScanNet [1] are dominated by classes like walls and floors, while many smaller categories occur far less frequently, making them harder to learn under standard training procedures. However, point clouds also introduce difficulties that have no direct analogue in grid-based 2D vision. Raw point clouds are inherently sparse, unordered, and irregularly distributed in 3D space, and their density can vary drastically across a single scan depending on sensor distance and occlusion [4]. Under a fixed preprocessing pipeline, a large fraction of the point budget is spent on already dominant regions, further compounding the class imbalance problem. These properties together make it insufficient to simply transfer 2D segmentation strategies to 3D, and present unique challenges that require dedicated solutions.

To address these challenges, substantial efforts have been devoted to multiple fronts. On the architecture side, methods have evolved from pioneering point-based networks such as PointNet and PointNet++ [4, 5], through local aggregation operators like PointConv [13] and graph-based designs such as DGCNN [14], to more expressive transformer-based architectures including Point Transformer [15] and Point Transformer V3 [11], steadily improving the capacity to capture both local geometric detail and long-range context. On the training side, loss functions such as weighted cross-entropy, focal loss [16], and Lovász-Softmax loss [17] have been widely adopted in point cloud segmentation to address class imbalance [18]. More recently, methods specifically designed for 3D settings have emerged, including multi-class imbalanced focus loss for long-tailed point distributions [19] and subspace prototype guidance that mitigates class bias

without additional inference cost [20]. Data augmentation techniques tailored to point clouds, including random rotation, scaling, and mix-based strategies, further help regularize training [21]. Beyond fully supervised settings, self-supervised pre-training methods such as PointContrast [22] and masked autoencoder approaches like Point-MAE [23] have shown that representations learned without labels can transfer effectively to downstream segmentation tasks.

In practice, raw point clouds are too large to be processed directly by most segmentation networks, making point sampling an indispensable preprocessing step [5, 24]. Standard strategies —such as grid sampling, random sampling, and farthest point sampling—select points based solely on geometric criteria, without considering the semantic content of different regions [4, 25]. As a result, classes that occupy larger spatial extents naturally retain more points after downsampling, while small or under-represented classes lose proportionally more of their already limited points, compounding the class imbalance before the network even begins to learn. This raises a natural question: *can point sampling itself become aware of semantic imbalance before segmentation begins?* Recent work has begun to treat sampling as a learnable component rather than a fixed heuristic. Learning to Sample [26] and differentiable point-cloud samplers [27] train a small network to produce task-optimised point subsets, while learned farthest point sampling [28] replaces the hand-crafted FPS rule with a differentiable approximation. However, existing learnable sampling methods have primarily focused on accelerating large-scale point cloud processing while preserving geometric fidelity, rather than exploiting the sampling stage to actively rebalance the semantic class distribution.

To investigate this question, we establish a unified Point Transformer V3 evaluation pipeline [11] on ScanNet [1], S3DIS [12], and PartNet [29]. We first measure how class frequency changes before and after standard grid sampling, and then compare these statistics with class-wise IoU. Across all three datasets, the analysis shows a consistent positive association between point frequency and segmentation quality, confirming that semantic imbalance is not only a property of the raw data but also a practical bottleneck after preprocessing.

Motivated by this finding, we study semantic-aware sampling as a way to intervene before feature extraction begins. The core idea is to reallocate a fixed point budget away from dominant structural regions and toward under-represented classes, while keeping the downstream segmentation backbone and evaluation protocol controlled. We first introduce an oracle sampler that uses ground-truth labels to test whether semantically informed budget allocation is useful at all. We then train a learnable Sampler that predicts point-wise sampling probabilities from point cloud features without access to test labels. The sampled subset is used to train or evaluate the downstream segmentor, and its sparse predictions are aggregated over the full point cloud through KNN-based test-time aggregation. This setup allows us to distinguish three effects that are often conflated: (1) whether semantic sampling produces a better sparse subset, (2) whether the downstream segmentor can exploit that subset, and (3) whether sparse predictions can be faithfully transferred back to the full scene.

Empirically, the sampler improves the quality of the selected sparse subset across datasets, while also exposing the remaining sparse-to-dense bottleneck. On ScanNet, the learned sampler improves Subset mIoU from 60.67 under Grid-Sampling to 64.78, close to the oracle sampler at 65.39, and improves full-scene KNN-3 mIoU from 58.88 to 61.04. On S3DIS and PartNet, the learned sampler gives smaller but consistent gains over Grid-Sampling, improving Subset mIoU from 56.03 to 58.61 and from 21.23 to 22.30 respectively. In contrast, the oracle sampler remains substantially stronger on S3DIS and PartNet, indicating that there is still unused headroom in semantic budget allocation. Overall, the results support the central hypothesis that sampling can improve the semantic composition of the retained points, but they also show that converting this sparse-subset advantage into full-scene accuracy depends critically on the downstream segmentor and the test-time aggregation strategy.

We summarize our contributions as follows:

- A systematic empirical analysis of class imbalance on ScanNet, S3DIS, and PartNet, showing that class frequency is consistently associated with class-wise segmentation quality.
- A semantic-aware sampling framework with both an oracle reference and a deployable learned Sampler, designed to test whether reallocating point budget at the sampling stage can improve balanced segmentation.
- A controlled evaluation of the sparse-to-dense prediction pipeline, including subset evaluation, KNN aggregation, stochastic multi-mask aggregation, and sampler/segmentor architecture ablations, showing that semantic sampling improves selected-subset quality while full-scene performance remains limited by test-time aggregation.

Chapter 2: Background

2.1 3D Point Cloud Representation

A point cloud is a set of samples taken from the surface or near-surface region of a 3D scene or object. Formally, a point cloud with N points can be written as

$$\mathcal{P} = \{(\mathbf{x}_i, \mathbf{f}_i)\}_{i=1}^{N}, \tag{1}$$

where $\mathbf{x}_i = (x_i, y_i, z_i)$ denotes the 3D coordinate of the i-th point and $\mathbf{f}_i$ denotes an optional d-dimensional attribute vector such as RGB colour, surface normal, or LiDAR intensity [3, 4]. Unlike a mesh, a raw point cloud does not explicitly store edges or faces. Unlike a voxel grid, it does not quantise space into a dense lattice. This makes point clouds lightweight and easy to acquire from sensors such as LiDAR and RGB-D cameras, but it also means that the data arrives as an unordered set rather than as a regular tensor.

Point Cloud differs from a 2D image in three fundamental ways. (1) *Unordered structure.* Unlike pixels arranged on a regular grid, point clouds have no canonical order: permuting the points does not change the scene. (2) *Non-uniform density.* Points are denser near observed surfaces and sparser in distant or occluded regions; the local density may vary significantly even within a single scan. (3) *Transformation sensitivity.* A point cloud is defined purely by coordinates, and the same scene may be captured under different rotations, translations, or scales across sensors or viewpoints.

These properties explain why a standard 2D CNN cannot be applied directly to raw point clouds. A 2D convolution assumes that data lie on a dense rectangular grid with fixed adjacency and translation-consistent kernels. Raw point clouds satisfy none of these assumptions. One may project a point cloud to 2D views or voxelise it into a 3D grid, but this introduces quantisation error, occlusion ambiguity, or high memory cost that grows rapidly with resolution [30, 31]. Modern point cloud networks either learn directly on unordered point sets or construct lightweight spatial structures that preserve geometric information without fully densifying the data.

2.2 Point Cloud Semantic Segmentation

Point cloud semantic segmentation is the task of assigning a semantic label to every point in an input cloud. Given a point set $\mathcal{P}$, the model predicts a label set $\mathcal{Y} = \{y_i\}_{i=1}^{N}$, where each y_i belongs to one of C semantic classes such as wall, floor, chair, or table. Compared with object classification, which produces one label for an entire shape, semantic segmentation requires the network to preserve fine local detail while also using scene-level context. This makes the task especially sensitive to sampling quality, local feature extraction, and class imbalance [4, 5].

The most widely reported metrics are mean Intersection over Union (mIoU) and mean class accuracy (mAcc). The main symbols used in these metrics are summarised in Table 1.

Table 1: Main symbols used in the semantic segmentation metrics.

Symbol	Definition
C	Number of semantic classes.
TP_c	Number of points of class c predicted correctly as class c.
FP_c	Number of points predicted as class c but belonging to other classes.
FN_c	Number of points belonging to class c but predicted as other classes.
IoU_c	Intersection over Union for class c.
Acc_c	Point-wise accuracy for class c.

For each class c, the class-wise IoU and accuracy are

$$IoU_c = \frac{TP_c}{TP_c + FP_c + FN_c}, \qquad Acc_c = \frac{TP_c}{TP_c + FN_c}. \tag{2}$$

The mean metrics are then computed as

$$mIoU = \frac{1}{C}\sum_{c=1}^{C} IoU_c, \qquad mAcc = \frac{1}{C}\sum_{c=1}^{C} Acc_c. \tag{3}$$

Intuitively, mIoU is the stricter metric because it penalises both false positives and false negatives, making it sensitive to over-segmentation and missed regions at the same time. mAcc can be read as the average per-class recall. It is useful when classes are imbalanced, because it prevents dominant classes from fully determining the final score.

Architecturally, this field has evolved through several well-established stages. PointNet was the first widely adopted network to process raw point sets directly. It handles unordered inputs by applying a shared MLP to each point and then using a symmetric aggregation function, typically max pooling, to obtain permutation-invariant features [4]. While elegant and effective, PointNet models each point largely independently before global pooling, so its ability to encode fine local structure is limited. PointNet++ addresses this limitation by introducing hierarchical set abstraction: it samples centroids, groups local neighbourhoods, and learns features from progressively larger regions [5]. This brings the point cloud pipeline closer to the multi-scale intuition of CNNs while still respecting irregular geometry.

Subsequent work strengthened local relation modelling in different ways. DGCNN dynamically builds a graph in feature space and applies EdgeConv to capture relations between a point and its neighbours, allowing the graph structure to evolve with the features during training [14]. PointConv, KPConv, and sparse convolution based methods offered other ways to approximate convolutional reasoning on irregular 3D data [13, 32, 31]. More recently, transformer-based models brought attention mechanisms to point clouds. Point Transformer introduced vector

attention that conditions feature aggregation on relative position, while later variants such as Point Transformer V2 and Point Transformer V3 improved scalability through more efficient attention and pooling designs [15, 33, 11]. Taken together, these developments suggest that point cloud semantic segmentation is already architecturally mature: the core challenge is no longer whether point clouds can be segmented effectively, but how to do so more efficiently, more robustly, and under harder data conditions such as imbalance.

2.3 Point Transformer V3

We introduce Point Transformer V3 (PTV3) [11] as the segmentation backbone for all experiments in this project. PTV3 achieves competitive segmentation quality while avoiding the computational overhead of costly neighbourhood construction and highly customised attention modules, through three tightly coupled architectural decisions: serialization, patch-based attention, and grid pooling.

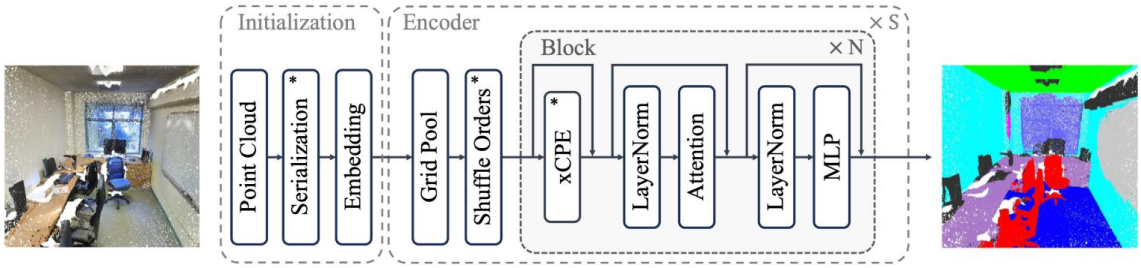


Figure 1: Overview of the PTV3 backbone. Points are first serialized into an ordered sequence, then processed through alternating attention blocks and grid pooling stages to produce hierarchical features.

Serialization replaces repeated neighbour search with a one-time mapping from 3D coordinates to a linear order, constructed so that spatially proximate points tend to be adjacent in the sequence. Local context is then approximated by attending within contiguous windows of this sequence, converting irregular spatial processing into a form amenable to hardware-efficient sequence operations. To prevent any single traversal order from introducing systematic bias, PTV3 rotates among multiple serialization orderings across attention blocks.

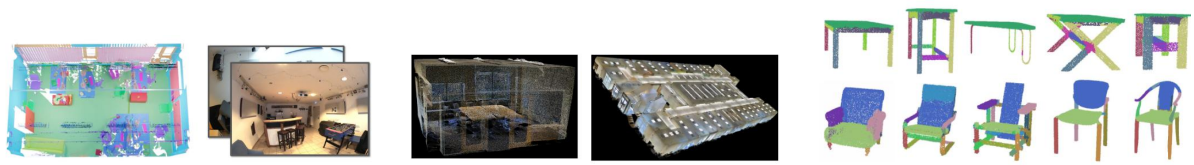
Within each serialized patch, attention is computed with an xCPE module inserted beforehand to inject local positional information. This allows the model to weight neighbours by both feature similarity and relative position, without the complex relative-position encodings used in earlier point transformers. The resulting block structure—positional encoding, windowed attention, residual connection, MLP—is deliberately simple, which contributes to its favourable scaling behaviour.

Between stages, *grid pooling* aggregates points within the same spatial cell into a single token, simultaneously reducing sequence length and expanding the receptive field. Unlike image

downsampling on a fixed lattice, this reduction is performed directly in 3D space, which keeps memory consumption tractable for large indoor scenes while preserving coarse geometric structure. Because serialization, attention, and pooling are tightly coupled, any change to the input point distribution propagates through the entire hierarchy rather than being absorbed at a single layer.

2.4 Datasets

Three indoor semantic segmentation benchmarks are used in this project: ScanNet, S3DIS, and PartNet. Representative examples are shown in Figure 2.



(a) Scannet scene examples (b) S3DIS scene examples (c) PartNet object examples

Figure 2: Representative examples of the three datasets used in this project.

ScanNet is a large-scale indoor RGB-D dataset collected by scanning real rooms with commodity sensors and reconstructing them into 3D scenes [1]. It contains 1,513 scans from 707 indoor spaces, annotated across 20 semantic classes. ScanNet reflects realistic clutter, occlusion, and sensor noise, making it well suited for evaluating scene-level robustness under naturalistic conditions.

Table 2: Datasets used in this project. ScanNet and S3DIS cover large-scale indoor scenes; PartNet targets fine-grained object part segmentation.

Dataset	**Task**	**Scale**	**Classes**	**Annotation**
ScanNet	Indoor scene seg.	1,513 scans, 707 spaces	20	Dense, point-wise
S3DIS	Indoor scene seg.	6 areas, ∼270 rooms	13	Dense, point-wise
PartNet	Object part seg.	26,671 shapes, 24 categories	Hierarchical	Hierarchical, part-wise

S3DIS (Stanford Large-Scale 3D Indoor Spaces) contains point cloud scans of large indoor areas including offices, classrooms, and meeting rooms [12]. The dataset covers six areas and approximately 270 rooms, with over 200 million points annotated into 13 semantic classes. Structural categories such as ceiling, floor, wall, table, and chair appear consistently across rooms, making S3DIS particularly valuable for cross-area generalisation experiments.

PartNet provides fine-grained part-level annotations for 3D objects rather than full indoor scenes [29, 34]. It contains 26,671 shapes from 24 object categories with hierarchical part annotations,

offering a complementary evaluation setting in which class imbalance manifests differently: some parts are large and spatially dominant, while others are small, thin, and semantically distinctive. Together with ScanNet and S3DIS, it allows the sampling strategies investigated in this report to be assessed across varying scene granularities.

2.5 Point Cloud Sampling

Point sampling is a standard preprocessing step in large-scale point cloud processing: raw scenes often contain far more points than a segmentation network can handle directly, and the choice of sampler controls not only computational cost but also which geometric structures and semantic regions are presented to the backbone.

Grid sampling partitions 3D space into uniform cells of side length v and retains one representative point per occupied cell, typically the centroid or a randomly chosen point within it [30, 31]. This regularises point density and integrates naturally with sparse convolution pipelines. However, the cell size v introduces a hard resolution limit: structures whose extent is smaller than v may collapse to a single cell, causing geometric detail to be irretrievably lost before the backbone is reached.

Random sampling draws a subset of size M uniformly at random from the N input points, either as a fixed subset or resampled each training iteration [24, 35]. The approach is trivially fast and serves as a mild regulariser when the sampled subset varies across epochs. Its weakness is the absence of any spatial coverage guarantee: the expected number of samples in a region is proportional to its point density, so already-dense surfaces are overrepresented and small structures may be missed entirely at low keep ratios.

Farthest point sampling (FPS) addresses this by constructing a subset $\mathcal{S} \subset \mathcal{P}$ that maximises the minimum pairwise distance between selected points. Concretely, starting from a randomly chosen seed p_1, each subsequent point is selected as

$$p_{k+1} = \underset{p \in \mathcal{P} \setminus \mathcal{S}_k}{\arg\max} \; \min_{p_j \in \mathcal{S}_k} \|p - p_j\|_2, \tag{4}$$

where $\mathcal{S}_k = \{p_1, \ldots, p_k\}$ is the set already selected [5, 25]. The resulting subset achieves approximately uniform spatial coverage and preserves global geometry better than random sampling. The cost, however, scales as $\mathcal{O}(NM)$ per iteration, which is prohibitive for large outdoor or indoor scenes where N can exceed 10^5 points. Moreover, FPS is driven purely by Euclidean distance: a large planar wall and a small semantically rare object may both be spatially covered, but no additional budget is allocated to the rare class.

These limitations have motivated *learnable sampling* methods that replace fixed geometric heuristics with trainable selection mechanisms. Learning to Sample and differentiable point-cloud samplers optimise a differentiable approximation to point selection so that the retained

subset preserves task-relevant information rather than merely geometric coverage [26, 27]. Learned FPS relaxes the hard greedy rule of Equation 4 into a soft, learned scoring function, making the process adaptive to the downstream objective [28]. Despite these advances, existing learnable samplers are trained primarily for geometric fidelity or overall task accuracy. None explicitly targets semantic class imbalance, leaving open the question of whether sampling can be guided by the distribution of semantic labels rather than by geometry alone.

2.6 Test-time Aggregation

Test-time aggregation is a necessary post-processing step when a segmentation network operates on a sampled subset rather than the full point cloud: the backbone predicts semantic scores only for the M selected points in $\mathcal{S}$, yet benchmark evaluation requires a label for every point in the original cloud $\mathcal{P}$. The choice of aggregation method determines how faithfully the sparse predictions transfer to unsampled regions, and matters most near object boundaries where the sampled subset may be sparse.

Nearest-neighbour assignment (KNN-1) is the simplest and most widely used aggregation rule. Each full point $\mathbf{x}_i \in \mathcal{P}$ inherits the predicted class-probability vector of its closest selected point,

$$\hat{\mathbf{p}}_i = \mathbf{p}_{j^*(i)}, \qquad j^*(i) = \underset{1 \le j \le M}{\arg\min} \|\mathbf{x}_i - \mathbf{s}_j\|_2, \tag{5}$$

and the predicted label is $\hat{y}_i = \arg\max_c \hat{p}_{i,c}$. The rule is deterministic and computationally negligible, but it partitions the scene into Voronoi cells around each selected point, producing piecewise-constant predictions that can misplace boundaries when neighbouring selected points belong to different classes.

K-NN interpolation softens this hard assignment by blending the predictions of the K nearest selected points using inverse-distance weights. Let $\mathcal{N}_K(i)$ denote the K nearest points in $\mathcal{S}$ to $\mathbf{x}_i$. The interpolation weight assigned to selected point $\mathbf{s}_j$ is

$$\alpha_{ij} = \frac{(\|\mathbf{x}_i - \mathbf{s}_j\|_2 + \varepsilon)^{-\gamma}}{\sum_{l \in \mathcal{N}_K(i)} (\|\mathbf{x}_i - \mathbf{s}_l\|_2 + \varepsilon)^{-\gamma}}, \qquad j \in \mathcal{N}_K(i), \tag{6}$$

where $\varepsilon > 0$ prevents division by zero and $\gamma > 0$ controls how sharply the nearest point dominates. The aggregated probability is then $\hat{\mathbf{p}}_i = \sum_{j \in \mathcal{N}_K(i)} \alpha_{ij} \mathbf{p}_j$, with $\hat{y}_i = \arg\max_c \hat{p}_{i,c}$. Setting $K = 1$ recovers nearest-neighbour assignment; larger K allows multiple selected points to jointly determine the prediction for each full point, which can smooth over erroneous individual predictions near object boundaries.

Chapter 3: Design and Implementation

This chapter describes the experimental pipeline used to test whether semantic-aware sampling can mitigate class imbalance in 3D point cloud segmentation. The central principle is to keep the downstream segmentation setting controlled while changing only how the input subset is selected. In this way, differences in class-wise IoU can be interpreted as consequences of the sampling strategy rather than as side effects of a different backbone, loss function, or evaluation protocol.

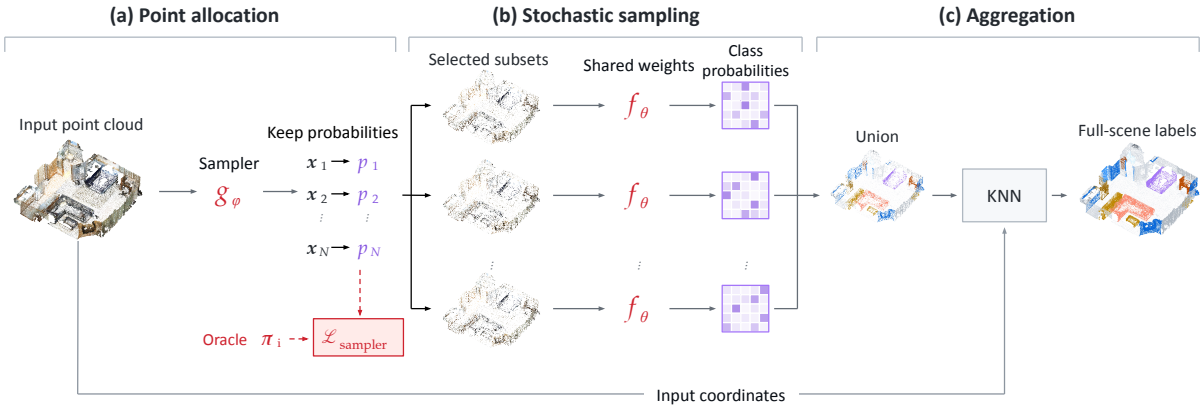


Figure 3: Overview of the proposed framework. **(a) Point allocation** (Section 3.3): the sampler g_ϕ predicts point-wise keep probabilities p_i, supervised during training by oracle targets π_i through $\mathcal{L}_{\text{sampler}}$. **(b) Stochastic sampling** (Sections 3.4 and 3.5): probabilities are calibrated to the target budget before independent masks are drawn; the shared segmentor f_θ predicts class probabilities on each selected subset. **(c) Aggregation** (Section 2.6): predictions are combined on the union of selected points, and KNN interpolation propagates class probabilities to the full point cloud.

3.1 Overall Framework

The proposed pipeline is illustrated in Figure 3. Given a point cloud

$$\mathcal{P} = \{(\mathbf{x}_i, \mathbf{a}_i, y_i)\}_{i=1}^{N}, \tag{7}$$

where $\mathbf{x}_i$ is the 3D coordinate, $\mathbf{a}_i$ denotes optional attributes such as colour or surface normal, and y_i is the semantic label available during training, the sampler selects a subset $\tilde{\mathcal{P}} \subset \mathcal{P}$ under a fixed point budget. A segmentation backbone f_θ then predicts semantic scores on $\tilde{\mathcal{P}}$. Since benchmark evaluation requires predictions on the original point cloud, the sparse predictions are finally aggregated over $\mathcal{P}$ using the test-time aggregation rules introduced in Section 2.6.

The key question is how $\tilde{\mathcal{P}}$ should be selected. Class-agnostic baselines such as grid sampling, random sampling, and farthest point sampling select points according to geometry or density. As a result, large structural classes tend to retain a larger share of the point budget, while rare

or small object classes may lose already limited evidence. The semantic-aware pipeline instead tries to assign more sampling opportunity to under-represented regions before the segmentation backbone is applied.

The full study uses three related settings. First, a grid-sampling baseline provides the class-agnostic reference. Second, an oracle sampler uses ground-truth labels to test the best-case effect of semantic budget reallocation. Third, a learnable Sampler predicts point-wise sampling probabilities from point cloud features without access to test labels. All three settings are evaluated under the same subset budget and the same sparse-to-dense aggregation protocol.

3.2 Oracle Adaptive Sampler

The oracle sampler is used to answer a diagnostic question: *if the sampler knew the semantic labels, would* reallocating point budget toward under-represented classes improve segmentation? For one training or validation sample, let

$$n_c = \sum_{i=1}^{N} \mathbf{1}(y_i = c), \qquad q_c = \frac{n_c}{N} \tag{8}$$

be the point count and point proportion of class c. Given a total budget of M selected points, the oracle sampler chooses a class cap τ such that

$$\sum_{c=1}^{C} \min(n_c, \tau) \approx M. \tag{9}$$

The number of retained points for class c is then

$$m_c = \min(n_c, \tau), \qquad r_c = \frac{m_c}{n_c}. \tag{10}$$

Dominant classes are truncated by the cap, while minority classes retain a larger fraction of their available points. Each point inherits the keep ratio of its ground-truth class,

$$\pi_i = r_{y_i}, \tag{11}$$

which can be interpreted as the oracle marginal probability that point i should be kept under the semantic budget rule.

This oracle sampler is not deployable because it uses labels at inference time. Its role is instead to provide a controlled upper-bound reference for the sampling policy. Importantly, it is an upper bound on class-aware budget allocation, not an absolute upper bound on final segmentation accuracy. A downstream segmentor may still require dense local context around boundaries or large structures, so a label-aware subset can improve balance while still losing information

needed for full-scene prediction.

3.3 Learnable Sampler

The learnable Sampler approximates the oracle marginal keep probability using only input point cloud features. It predicts

$$p_i = \sigma(g_\phi(\mathbf{x}_i, \mathbf{a}_i)), \qquad i = 1, \ldots, N, \tag{12}$$

where g_ϕ is a point cloud network followed by a scalar prediction head and σ is the sigmoid function. The main experiments use a compact point-cloud sampler architecture; Section 4.4 and Table 8 replace it with alternative backbones to test whether the conclusion depends on the sampler family.

During sampler training, the target is the oracle marginal probability π_i defined above. The sampler loss is a point-wise regression objective,

$$\mathcal{L}_{\text{sampler}} = \frac{1}{N} \sum_{i=1}^{N} (p_i - \pi_i)^2. \tag{13}$$

This objective makes the sampler learn a semantic preference over points rather than a purely geometric coverage rule. Ground-truth labels are used only to construct the training target; at test time, g_ϕ receives only point cloud coordinates and attributes.

3.4 Budget Calibration and Subset Generation

Raw probabilities from the sampler are not sufficient by themselves, because two different scenes may have different point counts and probability distributions. To make evaluation comparable to the baselines, the predicted probabilities are calibrated to match a target budget. We apply a scalar bias in logit space,

$$\bar{p}_i = \sigma(\text{logit}(p_i) + b), \tag{14}$$

where b is chosen so that

$$\sum_{i=1}^{N} \bar{p}_i \approx M. \tag{15}$$

This calibration preserves the relative ordering learned by the sampler while controlling the expected number of selected points. A stochastic mask is then drawn as

$$z_i \sim \text{Bernoulli}(\bar{p}_i), \tag{16}$$

and the selected subset is

$$\tilde{\mathcal{P}} = \{(\mathbf{x}_i, \mathbf{a}_i, y_i) : z_i = 1\}. \tag{17}$$

For deterministic reporting, the same budget and aggregation rules are used across all methods. For the scaling of the test-time allocation studied in Section 4.3, multiple independent masks can be sampled and combined into a union before full-scene aggregation.

3.5 Downstream Segmentor Training and Evaluation

After a selector has produced sparse subsets, the downstream segmentor is trained or evaluated on those selected points. The default segmentation backbone is Point Transformer V3, introduced in Section 2.3; Section 4.4 and Table 7 further ablate alternative downstream segmentor architectures. All segmentors are trained with the standard semantic segmentation objective

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Lovasz}}, \tag{18}$$

where cross-entropy provides point-wise supervision and Lovász-Softmax [17] aligns optimisation with IoU-oriented evaluation. When a learned sampler is used, the downstream segmentor can be retrained on the sampler-selected subsets so that the backbone sees the same type of sparse evidence during training and validation.

Three evaluation views are reported throughout the experiments. *Subset* mIoU is computed directly on the selected sparse points and measures the quality of the sampled evidence before reconstruction. *KNN-1* and *KNN-3* mIoU aggregate the sparse predictions over the full point cloud using nearest-neighbour assignment and three-neighbour inverse-distance interpolation respectively. These metrics separate the quality of the sampler from the difficulty of transferring sparse predictions back to the original dense scene.

3.6 Implementation Details

The implementation is built on the Pointcept codebase [36]. ScanNet and S3DIS use point coordinates together with colour and normal features; PartNet is object-centred and mainly relies on geometric coordinates. Dataset preprocessing follows the corresponding Point Transformer V3 settings, including voxelisation, cropping, and standard geometric or colour augmentation during training. Due to time and computational constraints, the experimental splits follow the efficient segmentation setting of Point Transformer V3 [11]: we use 10% subsets for ScanNet and PartNet, and a 20% subset for S3DIS. All experiments are conducted on NVIDIA RTX 3090 and NVIDIA RTX 5090 GPUs.

Unless otherwise stated, the learnable sampler is implemented with a lightweight PTV3-style point-cloud architecture, while the downstream segmentor uses the standard PTV3 semantic

segmentation backbone. This design keeps the sampler substantially smaller than the segmentor and makes the comparison focus on the allocation of the point budget rather than on backbone capacity.

The target sampling budget is dataset-specific and is matched across grid, oracle, and learned samplers within each dataset. For ScanNet and S3DIS, this budget is defined by standard grid sampling with voxel size 0.04, denoted as *Grid004*. For PartNet, which is object-centred and uses normalised object coordinates, the budget follows the PartNet fixed-subset reference rather than the scene-level Grid004 convention. Thus the comparison changes which points are retained rather than how many points are retained within each dataset.

Stage 1: Oracle-selected segmentor training. We first train a downstream segmentor on oracle-selected subsets under the dataset-specific target budget. The oracle uses ground-truth labels only during training and analysis, and therefore provides a stable reference for what class-aware budget allocation can achieve when label information is available.

Stage 2: Learnable sampler training. We then train the learnable sampler to approximate the oracle marginal keep probabilities from input point features only. The sampler has no access to test labels; its supervision is the oracle marginal probability computed from the training split. This stage isolates whether a compact neural sampler can learn the label-aware allocation pattern without directly optimising the downstream segmentation network.

Stage 3: Sampler-index segmentor retraining. Finally, we materialise sampler-selected indices, calibrated to the same dataset-specific budget, and retrain the downstream segmentor on those selected subsets. When applicable, the segmentor is initialised from the oracle-selected checkpoint from Stage 1, so the comparison focuses on the effect of replacing oracle indices with learned sampler indices. The default hyperparameters for all three stages are summarised in Appendix Table 9.

A more principled alternative would be to jointly optimise the sampler and segmentor with the downstream segmentation loss. We leave this end-to-end formulation for future work: under the available time and compute budget, the staged heuristic provides a more stable and controlled way to study whether semantic-aware point allocation is beneficial.

All compared methods are matched by sampling budget within each dataset. The same downstream backbone family, segmentation loss, and validation protocol are used when comparing grid sampling, oracle sampling, and learned sampling. This controlled setting is essential: the purpose of the experiments is not to claim a new strongest segmentation backbone, but to isolate how the choice of sampler changes the semantic evidence available to the backbone and how much of that evidence survives the sparse-to-dense aggregation step.

Chapter 4: Results and Discussion

This chapter evaluates whether semantic-aware sampling can improve class-balanced point cloud segmentation under matched sampling budgets. Section 4.1 first examines class distributions and their relationship with segmentation quality on ScanNet, S3DIS, and PartNet. Section 4.2 then compares grid sampling, oracle semantic-aware sampling, and the learned sampler across the three datasets. Section 4.3 analyses where the sampler helps and where it loses performance, including class-wise tradeoffs, scaling of the test-time allocation, aggregation choices, and inference cost. Finally, Section 4.4 tests whether the conclusions depend on the sampler or segmentor architecture, and Sections 4.5–4.6 summarise the implications and remaining limitations.

4.1 Exploratory Statistics

We examine class distributions and their relationship with segmentation quality on ScanNet, S3DIS, and PartNet. Figure 4 shows the mean number of raw points per validation sample for each class, sorted in descending order on a logarithmic vertical axis. On ScanNet, structural classes such as *wall* (id 0) and *floor* (id 1) dominate the total point count, while object-level classes such as *picture* (id 10) and *sink* (id 17) occupy far smaller proportions. S3DIS shows the same pattern, with *ceiling* (id 0), *floor* (id 1), and *wall* (id 2) accounting for the majority of points. PartNet also exhibits a long-tailed distribution across its unified part labels.

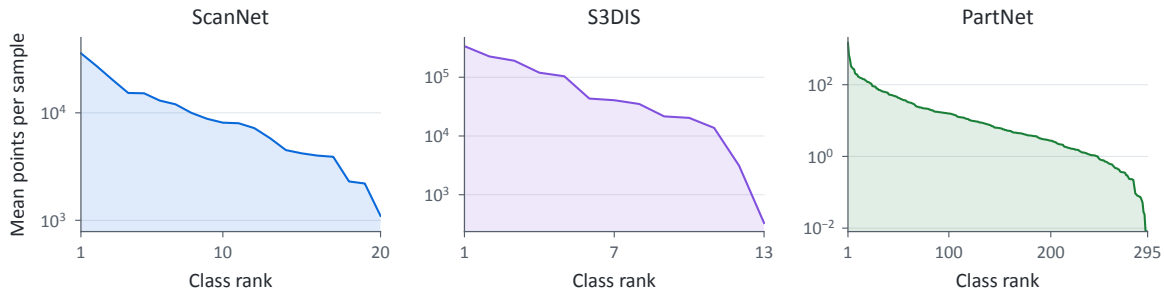


Figure 4: Class distributions in the raw validation sets. Classes are ranked by their mean number of points per validation sample, shown on a logarithmic vertical axis without normalisation. All classes are included: 20 for ScanNet, 13 for S3DIS, and 295 for PartNet.

Figure 5 plots class frequency against class-wise IoU for the reproduced grid baselines. All three datasets exhibit a positive association between class frequency and per-class IoU. Linear fits in log-frequency space yield $R^2 = 0.21$ for ScanNet, 0.61 for S3DIS, and 0.34 for PartNet, indicating that the strength of the association varies across datasets. Together, the frequency skew and the observed performance association motivate investigating semantic-aware allocation at the sampling stage.

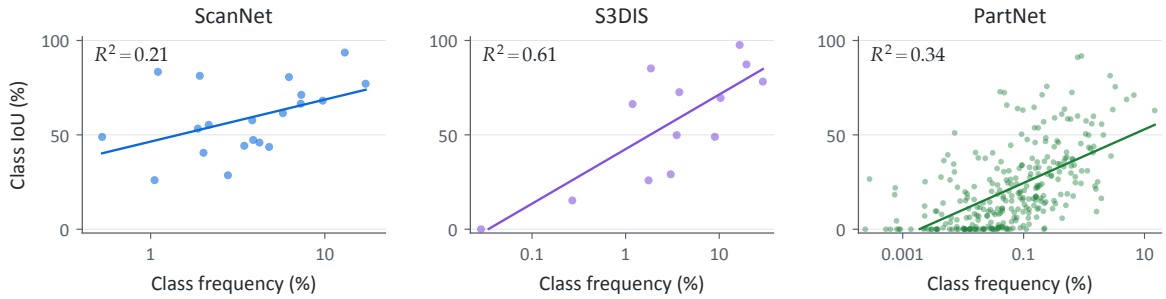


Figure 5: Class frequency and segmentation quality under the grid-sampling baselines. Each point represents one class; frequency is its share of validation points, shown on a logarithmic axis. Lines are linear fits in log-frequency space, with R^2 reported in each panel.

4.2 Main Results

Table 3 summarises validation results across the three evaluation datasets. We report up to three metrics per setting: Full mIoU (KNN-1) aggregates selected-point predictions over the full point cloud via nearest-neighbour assignment; Subset mIoU is computed directly on the selected sparse points; and Full mIoU (KNN-3) uses three-neighbour inverse-distance weighted interpolation for full-scene prediction. The distinction between sparse-subset evaluation and KNN-based full-scene aggregation is introduced in Section 2.6. On ScanNet, the learned sampler with downstream retraining and scaling of the test-time allocation achieves 60.90 Full mIoU (KNN-1), 64.78 Subset mIoU, and 61.04 Full mIoU (KNN-3). On S3DIS, the oracle adaptive sampler remains much stronger, while the learned sampler provides a moderate improvement over the grid-sampling baseline. On PartNet, the learned sampler gives a modest improvement over grid sampling, while the oracle sampler remains stronger.

Table 4 presents a detailed class-wise comparison on ScanNet in the controlled setting where class imbalance is most directly visible. Three systems are compared under identical backbone, loss, and optimisation settings: the Grid-Sampling baseline, an Oracle Adaptive sampler with access to ground-truth labels at inference time, and the learned Sampler proposed in Section 3.2.

The class-wise patterns in Table 4 are further analyzed in Section 4.3.

4.3 Analysis

Tradeoff of the sampler. The class-wise comparison in Table 4 shows that the sampler's gains are not uniform across classes, reflecting a trade-off between recovering point evidence for rare object classes and preserving dense contextual support for large structural ones. Relative to Grid-Sampling, the Sampler improves IoU on 17 of 20 classes, with the largest gains on shower curtain (+11.9), counter (+3.9), picture (+3.5), toilet (+3.5), bathtub (+3.3), table (+3.2), desk (+3.1), and cabinet (+3.1) percentage points. These are classes whose point evidence is easily diluted when a geometry-only sampler allocates a disproportionate share of the point budget to dominant

Table 3: Main validation results (mIoU, %) across datasets and aggregation settings. Subset evaluates predictions directly on selected sparse points, KNN-1 uses nearest-neighbour aggregation over the full point cloud, and KNN-3 uses three-neighbour inverse-distance weighted aggregation, following the test-time aggregation protocol in Section 2.6. The shaded column highlights our method; **bold** marks the best result in each row.

Dataset	Aggregation	Grid-Sampling	Oracle Sampling	Sampler (ours)
ScanNet	Subset	60.67	**65.39**	64.78
	KNN-1	58.64	**65.12**	60.90
	KNN-3	58.88	**65.16**	61.04
S3DIS	Subset	56.03	**71.12**	58.61
	KNN-1	56.42	**68.70**	58.00
	KNN-3	56.56	**68.92**	58.22
PartNet	Subset	21.23	**24.96**	22.30
	KNN-1	21.30	**23.39**	22.16
	KNN-3	21.30	**23.48**	22.17

structural regions, leaving insufficient points to characterise smaller or sparser objects. The primary degradations occur on refrigerator (-3.6) and door (-1.2), where recognition depends more heavily on boundary continuity and surrounding context respectively.

Figure 6(a) relates the selected union point count of each ScanNet class to its KNN-3 IoU gain over Grid-Sampling. Across the 20 classes, the Pearson correlation between log selected point count and IoU gain is $r = -0.32$; classes with more selected points do not consistently obtain larger gains. For example, shower curtain gains 11.9 percentage points with about 0.10M selected points, whereas wall and floor have about 7.61M and 6.96M selected points but gain only 0.1 and 1.1 points, respectively. Door and refrigerator remain below the baseline despite retaining about 1.08M and 0.15M points. These comparisons suggest that selected point count alone does not explain the variation in class-wise gains. Figure 6(b) reports the fraction of raw points retained in the selected union for each class. These keep ratios range from 59.5% for window to 77.4% for floor; shower curtain retains 63.4% despite achieving the largest IoU gain.

The Sampler also outperforms the oracle adaptive sampler on several classes, including curtain ($+8.7$), bookshelf ($+6.0$), and bed ($+5.1$). This does not imply that the learned sampler leverages superior semantic information compared with the oracle. Rather, the oracle serves as an upper bound on class-aware budget allocation, not on segmentation accuracy per se: it reallocates points according to ground-truth labels but does not optimise the downstream segmentor's spatial context. The learned sampler can therefore retain geometrically useful context that is not captured by the oracle's purely label-balanced allocation, so classes that benefit from context can outperform the oracle, whereas rare or thin classes such as picture, door, and window remain substantially below it. The result clarifies the core trade-off: semantic-aware sampling

Table 4: ScanNet class-wise comparison (mIoU and mAcc in %). Red indicates improvement over Grid-Sampling, blue indicates degradation; colour intensity reflects magnitude.

ID	Class	Point Count ($\times 10^3$)	Grid-Sampling		Oracle Sampling		Sampler (ours)	
			mIoU	mAcc	mIoU	mAcc	mIoU	mAcc
0	wall	35.8	77.1	92.8	**80.1**	90.3	77.2	93.8
1	floor	27.2	93.6	97.4	94.8	97.9	94.7	98.3
2	cabinet	10.0	43.6	62.0	**48.9**	64.4	**46.7**	**57.8**
3	bed	15.3	71.2	77.4	**65.8**	**73.6**	70.9	76.7
4	chair	13.0	80.6	91.0	82.4	92.5	82.7	90.9
5	sofa	12.0	61.5	74.9	**67.6**	79.2	64.3	**80.2**
6	table	8.0	57.7	73.0	**64.6**	**80.0**	**60.9**	73.9
7	door	7.2	44.2	57.9	**63.7**	**81.5**	43.0	**54.1**
8	window	8.8	45.9	58.7	**63.8**	**80.1**	46.9	**61.9**
9	bookshelf	20.3	68.1	77.1	**64.5**	**82.8**	70.5	**81.5**
10	picture	2.2	26.0	32.1	**51.2**	**59.8**	**29.5**	**37.2**
11	counter	3.9	53.3	60.0	**61.9**	**70.7**	**57.2**	**64.8**
12	desk	8.1	47.3	66.7	**56.1**	**75.4**	**50.4**	68.9
13	curtain	15.2	66.5	74.4	**60.4**	72.0	**69.1**	**77.3**
14	refrigerator	4.2	40.5	44.8	**47.0**	**56.9**	**36.9**	**40.1**
15	shower curtain	4.5	55.3	62.2	**72.6**	**86.2**	**67.2**	**78.2**
16	toilet	2.3	83.4	92.7	**89.3**	**98.6**	**86.9**	94.9
17	sink	1.1	48.9	56.5	**55.8**	**66.0**	50.0	57.2
18	bathtub	4.0	81.3	85.5	84.8	**92.2**	84.6	**88.9**
19	otherfurniture	5.8	28.6	34.2	**38.9**	**49.0**	**31.3**	**37.4**
Avg		10.4	58.7	68.6	**65.7**	**77.5**	61.0	70.7

improves per-class evidence for underrepresented objects, but full-scene accuracy still depends on retaining sufficient local geometry for large or context-sensitive classes.

Scaling of the test-time allocation. We next ask *whether the learned sampler can be made stronger at inference time without retraining*. The idea is simple: *instead of sampling once, we sample N independent masks, merge their selected points, and aggregate predictions from this larger selected set back to the full point cloud*. Figure 7(a) shows that this mainly benefits full-scene aggregation. From $N = 1$ to $N = 32$, KNN-1 mIoU increases from 59.67 to 60.90, and KNN-3 mIoU increases from 59.91 to 61.04. Most of the gain arrives by $N = 16$; adding more masks after that gives only marginal improvement. The tradeoff is that Subset mIoU decreases from 64.78 to 61.91, because the merged selected set becomes less selective as more points are added. The coverage statistics in Table 5 show the same pattern: raw-point coverage rises quickly up to $N = 16$ and then saturates.

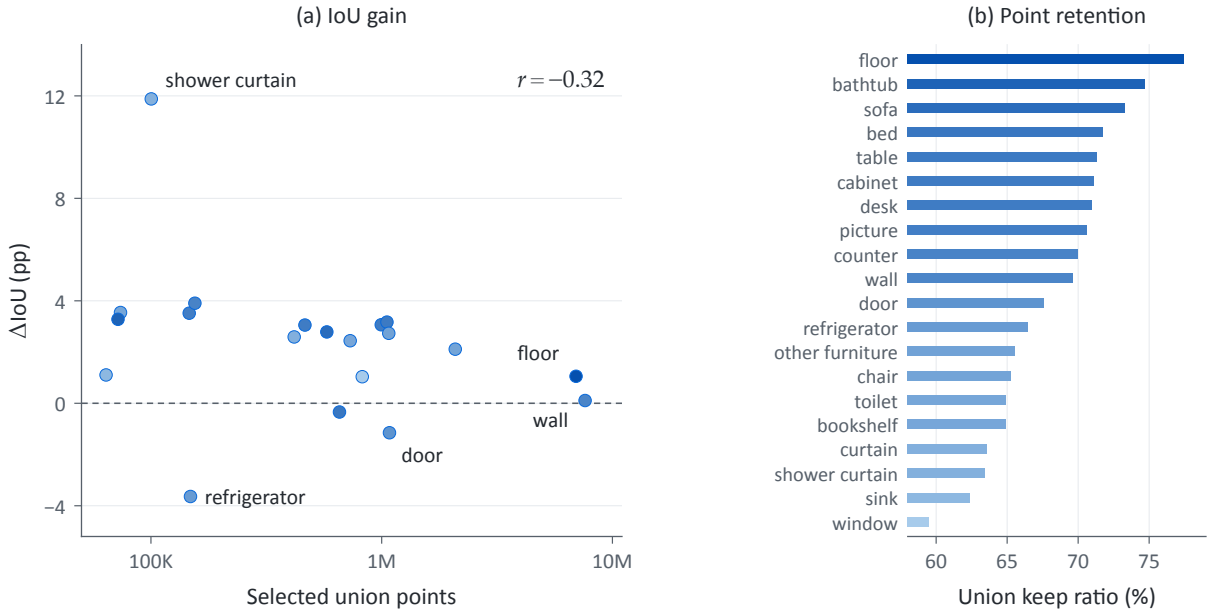


Figure 6: Class-wise sampling and segmentation gain on ScanNet. (a) Selected union point count (log scale) versus KNN-3 IoU gain over Grid-Sampling in percentage points. Each point is one class; the dashed line denotes zero gain and r is the Pearson correlation between log point count and IoU gain. (b) Union keep ratio, defined as selected union points divided by raw points for each class, sorted in descending order. Darker blue indicates a higher keep ratio.

Table 5: Coverage growth under stochastic test-time allocation on ScanNet. The union covers more raw points as independent masks are accumulated, but coverage saturates after roughly 16 masks.

Masks N	1	2	4	8	16	32
Union / raw (%)	23.01	35.68	49.71	61.13	67.06	68.73

Different aggregation settings. Test-time aggregation over multiple stochastic masks further improves full-scene mIoU, with most of the gain coming from replacing nearest-neighbour aggregation with inverse-distance weighted interpolation. On ScanNet, we fix the learned sampler and downstream PTv3 segmentor, draw 24 stochastic masks, and aggregate predictions on the union of selected points (covering 68.3% of all raw points). We use probability aggregation for all settings, because it gives a direct convex combination of class distributions after softmax and keeps the comparison focused on the interpolation neighbourhood. Increasing the interpolation neighbourhood is beneficial but saturates quickly: compared with the $K = 1$ nearest-neighbour baseline, probability aggregation improves full-scene mIoU by 0.12, 0.16, and 0.18 points for $K = 3$, $K = 8$, and $K = 16$, respectively. Figure 7(b) summarises this probability-based aggregation comparison.

Params and speed. Table 6 reports the model size and single-mask forward speed of the learned sampler pipeline. The sampler is lightweight relative to the downstream PTv3 segmen-

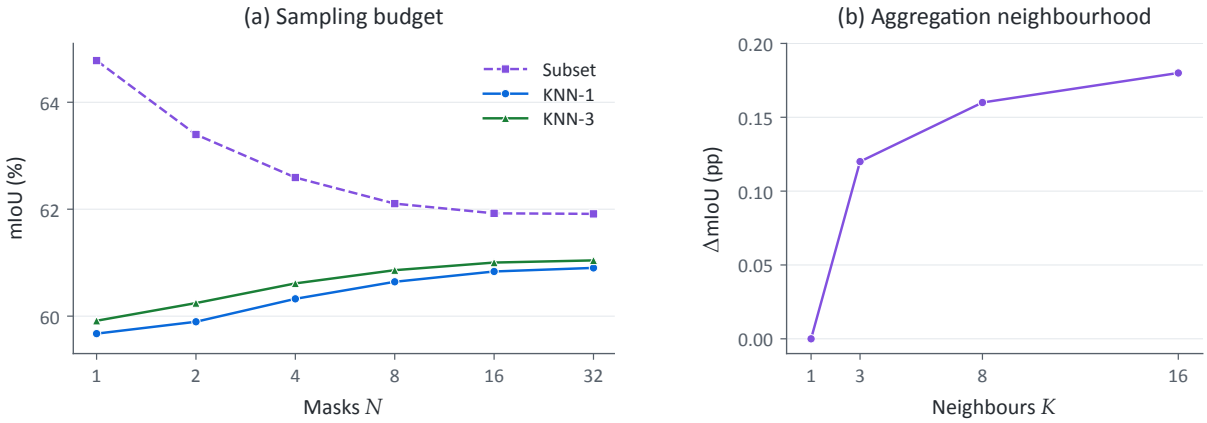


Figure 7: Test-time sampling and aggregation ablations on ScanNet with the learned sampler and PTv3 segmentor fixed. (a) mIoU as the number of stochastic masks N increases. Subset (dashed) is evaluated on selected points; KNN-1 and KNN-3 are evaluated on all raw points using nearest-neighbour and three-neighbour inverse-distance aggregation, respectively. (b) Full-scene mIoU gain over $K = 1$ as the interpolation neighbourhood increases, using probability aggregation with 24 masks.

tor: the ScanNet and S3DIS samplers contain only 2.69M parameters, and the smaller PartNet sampler contains 1.20M parameters, compared with approximately 46.17M parameters for the segmentor. When only neural network forward passes are counted, the sampler is not the runtime bottleneck: the downstream segmentor forward is slightly slower on all three datasets despite operating on the selected subset. For large S3DIS rooms, the KNN-3 aggregation step also becomes a visible cost because logits must be propagated back to over one million raw points per scene on average. These timings exclude stochastic mask generation, calibrated Bernoulli selection, and final grid cleanup, isolating the model-forward cost that scales with the sampler and segmentor architectures. The scaling of the test-time allocation in Section 4.3 trades repeated forward passes for improved full-scene accuracy.

Table 6: Parameter count and single-mask model-forward speed of the learned sampler pipeline. Parameter counts are computed from trained checkpoints. Timing is averaged over the validation split on one GPU of the autodl-5090 server, with CUDA synchronisation around each stage. Model loading and dataset construction are excluded. Sampler forward reports only the learned sampler network pass on raw points; calibrated Bernoulli selection and grid cleanup are excluded. Segmentor forward reports the sparse PTv3 network pass on selected points; KNN-3 aggregation maps sparse logits back to the full point cloud.

Dataset	Scenes	Sampler params (M)	Segmentor params (M)	Sampler forward (s)	Segmentor forward (s)	Aggregation (s)	Total (s/scene)
ScanNet	312	2.69	46.17	0.023	0.035	0.007	0.066
S3DIS	68	2.69	46.17	0.036	0.042	0.078	0.156
PartNet	2619	1.20	46.18	0.018	0.031	0.001	0.049

4.4 Ablations

We conduct two architecture ablations on ScanNet to assess the generality of the proposed pipeline. The first holds the sampling pipeline fixed and varies the downstream segmentor backbone; the second holds the downstream segmentor fixed and varies the sampler backbone. All tables report validation mIoU (%) under three evaluation settings: KNN-1 (nearest-neighbour aggregation to all raw points), KNN-3 (three-neighbour inverse-distance weighted aggregation), and Subset (direct evaluation on the selected sparse points without full-scene aggregation). These settings follow the test-time aggregation definitions in Section 2.6.

Segmentor architecture. Table 7 examines whether the proposed sampler generalises across downstream segmentor architectures, replacing PTv3 with two SpUNet variants while keeping the sampling pipeline unchanged. The learned sampler with downstream retraining transfers across segmentor families, consistently improving Subset mIoU over the fixed-oracle selector, though the gap on KNN-1 and KNN-3 persists across both backbones. Thus, the method is not architecture-specific, while the downstream backbone capacity remains the primary factor limiting the conversion of sparse-subset gains into full-scene accuracy.

Table 7: Segmentor architecture ablation on ScanNet. The learned sampler transfers across architectures, improving Subset mIoU in both SpUNet variants, though KNN-1 and KNN-3 full-scene mIoU remain below the fixed-oracle selector. The sampling pipeline is fixed; only the downstream segmentor backbone is varied.

Segmentor	Selector / training	KNN-1	Subset	KNN-3
SpUNet-small	Oracle Sampler	60.06	61.60	60.14
	Learned Sampler	58.23	62.12	58.43
SpUNet-base	Oracle Sampler	62.53	63.79	62.63
	Learned Sampler	60.33	64.41	60.54

Sampler architecture. Table 8 ablates the sampler architecture by replacing the PTv3-style sampler with SpUNet-small and SpUNet-base samplers, while keeping the downstream PTv3 segmentor fixed. The downstream settings follow the staged protocol in Section 3.6: *No segmentor retraining* evaluates a new sampler with the existing downstream PTv3 segmentor from the default pipeline, while *Segmentor retrained* corresponds to Stage 3, where PTv3 is trained on the subsets selected by that sampler. The non-PTv3 samplers achieve competitive KNN-1 and KNN-3 mIoU after segmentor retraining on sampler-selected subsets, and increasing sampler capacity from SpUNet-small to SpUNet-base primarily raises Subset mIoU. However, neither SpUNet variant improves KNN-1 or KNN-3 mIoU over the PTv3-style sampler, suggesting that the primary limitation lies in aggregating sparse selected evidence into dense full-scene predictions rather than in sampler backbone capacity.

Table 8: Sampler architecture ablation on ScanNet. The two downstream settings follow the staged training protocol in Section 3.6: *No segmentor retraining* keeps the downstream PTv3 fixed, while *Segmentor retrained* trains PTv3 on each sampler's selected subsets. Sampler backbone capacity has limited impact on KNN-1 and KNN-3 mIoU after retraining, suggesting that the aggregation step is the primary bottleneck.

Sampler	**Downstream setting**	**KNN-1**	**Subset**	**KNN-3**
PTv3-style	Segmentor retrained	59.72	64.90	59.95
SpUNet-small	No segmentor retraining	56.23	61.67	56.47
	Segmentor retrained	58.89	64.24	59.11
SpUNet-base	No segmentor retraining	56.71	62.80	56.96
	Segmentor retrained	59.12	65.16	59.35

4.5 Discussion

The experiments support three main observations. First, semantic-aware sampling does change the evidence available to the segmentation backbone. The oracle sampler improves results substantially, showing that reallocating point budget according to semantic class is useful in principle. The learned Sampler recovers part of this effect without using test labels, especially at the subset level, where the selected sparse points are evaluated directly. This confirms that the sampling stage can be used as an intervention point for class imbalance rather than only as a computational shortcut.

Second, the gap between Subset mIoU and KNN mIoU is central to interpreting the method. A high-quality selected subset does not automatically produce an equally strong full-scene prediction, because unsampled points still need to inherit semantic scores from nearby selected points. KNN interpolation and stochastic multi-mask aggregation reduce this gap by increasing coverage and smoothing local prediction noise, but the improvement eventually saturates. As more masks are accumulated, the union covers more of the raw scene, yet it also becomes less selective; therefore, subset performance can decrease even when full-scene coverage increases. This behaviour indicates that the main limitation is not only whether the sampler chooses useful points, but also whether sparse evidence can be transferred back to the full point cloud faithfully.

Third, the oracle sampler should be interpreted carefully. It is an upper-bound reference for semantic budget allocation, not a guarantee of the best possible segmentation accuracy. Some classes require dense local context or boundary continuity, and a label-balanced subset can still remove information that the downstream segmentor needs. This explains why the learned sampler can be competitive with, or occasionally better than, the oracle on some context-heavy classes, while still trailing it on rare or thin structures. The architecture ablations reinforce the same conclusion: changing the sampler or segmentor backbone affects the numbers, but the persistent Subset–KNN gap suggests that the sparse-to-dense transfer step is a major bottleneck.

4.6 Limitations

The current approach has several limitations. The most important one is the sparse-to-dense prediction bottleneck. The sampler and segmentor operate on a selected subset, but the final benchmark is evaluated on a denser point cloud. Nearest-neighbour and inverse-distance KNN aggregation are simple and efficient, yet they cannot fully reconstruct semantic detail that was not present in the selected subset, especially around thin structures and object boundaries.

The second limitation is that semantic reallocation can reduce useful context for some classes. Classes such as doors, refrigerators, and other boundary-dependent objects may require surrounding geometry to be recognised reliably. A sampler that improves class balance may still remove contextual points that are useful for these categories. This creates a real trade-off between minority-class recovery and local geometric continuity.

The third limitation is computational. Multi-mask aggregation improves full-scene results by covering more points, but it requires repeated sampler and segmentor passes. This is acceptable for analysis but less attractive for real-time or resource-constrained settings.

Another limitation is the staged training procedure. In principle, the sampler and segmentor should be optimised jointly with the downstream segmentation objective, so that point allocation is directly shaped by the final task loss. In this project, we instead use a three-stage heuristic training pipeline for stability and experimental control: oracle-selected segmentor training, sampler supervision from oracle marginal probabilities, and downstream retraining on sampler-selected subsets. This decomposition produces positive results, but it leaves room for future work on end-to-end sampler–segmentor co-training. Finally, the experiments are conducted on standard benchmarks with fixed label sets. More diverse scenes, different sensor modalities, and open-vocabulary settings may expose additional failure modes not covered by the current evaluation.

Chapter 5: Conclusion and Further Work

5.1 Conclusion

This project investigated whether semantic class imbalance in 3D point cloud segmentation can be addressed at the sampling stage rather than only through loss reweighting or backbone design. A controlled Point Transformer V3 based pipeline was established on ScanNet, S3DIS, and PartNet, and the exploratory statistics showed a consistent positive association between class frequency and class-wise IoU. This confirms that under-represented categories are not merely rare in the dataset; they are also systematically harder to preserve and segment after standard preprocessing.

Motivated by this observation, the project evaluated semantic-aware sampling through both an oracle reference and a learnable Sampler. The oracle sampler showed that reallocating a fixed point budget according to semantic labels can substantially improve balanced segmentation quality. The learned Sampler recovered part of this effect without using test labels, improving the selected sparse subset and several minority or small-object classes. These results support the main hypothesis of the project: sampling is not a neutral preprocessing step, and changing the sampler changes the semantic evidence available to the segmentation backbone.

At the same time, the experiments also show that subset-level improvement is not equivalent to full-scene improvement. KNN aggregation, probability-based aggregation, and scaling of the test-time allocation all help transfer sparse predictions back to the full point cloud, but the gain saturates. The persistent gap between Subset mIoU and full-scene KNN mIoU indicates that sparse-to-dense aggregation is the main bottleneck of the current pipeline. Therefore, the most defensible conclusion is not that semantic-aware sampling alone solves class imbalance, but that it exposes a useful new intervention point whose benefit depends strongly on the quality of the downstream aggregation stage.

5.2 Reflection

A recurring difficulty in this project was separating the effect of the sampler from the effect of the downstream segmentor and evaluation protocol. Early experiments showed that small changes in budget matching, validation aggregation, or segmentor retraining could change the interpretation of a result. This made a controlled experimental design essential: the backbone, budget, loss, and aggregation rules had to be matched before class-wise differences could be attributed to the sampling strategy itself.

Another lesson is that partial improvements can still be informative. The learned Sampler does not fully close the gap to the oracle sampler, and its subset gains do not always translate into full-scene gains. However, these limitations identify where the current system fails: not only in

estimating which points should be sampled, but also in propagating sparse predictions back to unsampled regions. This turns a negative or incomplete result into a clearer research direction.

The project also highlighted the cost of large-scale empirical work. Training samplers, retraining segmentors, and evaluating multiple aggregation settings requires substantial GPU time. Prioritising controlled comparisons over unbounded hyperparameter sweeps was necessary both for resource efficiency and for making the final conclusions interpretable.

5.3 Further Work

The most direct direction for future work is to improve sparse-to-dense aggregation. KNN interpolation is simple and reliable, but it ignores richer local geometry, confidence, and class-boundary structure. A learned aggregation module or a boundary-aware aggregation rule may convert subset-level gains into stronger full-scene performance.

A second direction is context-aware sampling. The current semantic reallocation mainly considers class balance, but some categories require more surrounding context than others. Future samplers could combine semantic rarity with geometric signals such as boundary density, local curvature, or neighbourhood uncertainty, so that rare classes receive additional evidence without removing too much context from structure-dependent classes.

Finally, efficient multi-mask inference remains an open problem. Stochastic mask aggregation improves coverage, but repeated forward passes increase runtime. Designing samplers that provide multi-mask-like coverage in a single pass, or distilling multi-mask predictions into a faster deterministic policy, would make semantic-aware sampling more practical for larger scenes and real-time systems.

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Appendices

Disclaimer

This report is submitted as part requirement for the undergraduate degree programme at Queen Mary University of London, and Beijing University of Posts and Telecommunications. It is the product of my own labour except where indicated in the text. The report may be freely copied and distributed provided the source is acknowledged.

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Acknowledgement of using Generative AI

I acknowledge the use of the following Generative AI tools

Publisher	Name, version, URL	Use	Section
Google	Google Gemini (latest version) https://gemini.google.com/	To help brainstorm ideas, search for relevant materials, and collect references.	Chapter 1-3
OpenAI	OpenAI GPT-5.4 https://openai.com/	To help search for relevant materials and collect references.	Chapter 1-2
OpenAI	Codex GPT-5.4 https://openai.com/	To assist with polishing the English expression and checking the logical flow of the report.	Chapter 1-5
Anthropic	Claude Opus 4.6 https://www.anthropic.com/claude	To assist with polishing the English expression and checking the logical flow of the report.	Chapter 1-5

I confirm the submitted work is my own work.

北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 1 - Supervisor

论文题目 Project Title	Multi-scale 3D object semantic segmentation based on Deep Learning		
题目分类 Scope	Implementation	Multimedia and Vision	Simulation
主要内容 Project description	Semantic segmentation of 3D point clouds is the process of assigning a class label to every point in a 3D dataset. This capability is crucial for a growing number of high-impact domains, including autonomous navigation for self-driving cars, sophisticated robotic interaction with complex environments, and the creation of immersive augmented reality experiences. However, processing raw 3D point cloud data presents unique and significant challenges compared to 2D images. This data is inherently unstructured, sparse, and irregular, lacking the grid-like pixel arrangement that allows for the direct application of conventional deep learning techniques like CNNs.This project will directly address these challenges by undertaking a comprehensive exploration of advanced deep learning methodologies. The primary goal is to develop a novel architecture to achieve better performance. The project will commence with a rigorous investigation of diverse and cutting-edge architectural paradigms to establish strong performance baselines across various datasets. Drawing insights from this analysis, a new network architecture will be designed and implemented, focusing on the intelligent fusion of multi-scale features to enhance performance. To display its applicability directly, the project will culminate in the development and validation of a functional 3D point cloud segmentation application demo.		
关键词 Keywords	Deep Learning, 3D Semantic segmentation		
主要任务 Main tasks	1 Conduct a rigorous investigation of diverse architectural paradigms to establish performance baselines		
	2 Establish an estimation system to calculate the metrics of an architecture		
	3 Using insights from this analysis, design and implement a novel network architecture that intelligently fuses multi-scale features for enhanced performance.		
	4 Build and validate a 3D point cloud segmentation demo to display the applicability clearly.		
主要成果 Measurable outcomes	1 A baseline neural network architecture for 3D semantic segmentation		
	2 A novel semantic segmentation algorithm that fuses multi-scale features		
	3 An end-to-end 3D point cloud segmentation demo application with intuitive visualization of the segmentation results		

北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 2 - Student

<table>
<tr><td>学院
School</td><td>International School</td><td>专业
Programme</td><td colspan="3">Internet of Things Engineering</td></tr>
<tr><td>姓
Family name</td><td>Zhao</td><td>名
First Name</td><td colspan="3">Jian</td></tr>
<tr><td>BUPT 学号
BUPT number</td><td>2022213320</td><td>QM 学号
QM number</td><td>221167928</td><td>班级
Class</td><td>2022215111</td></tr>
<tr><td>论文题目
Project Title</td><td colspan="5">Multi-scale 3D object semantic segmentation based on Deep Learning</td></tr>
<tr><td>论文概述
Project outline

Write about 500-800 words

Please refer to Project Student Handbook section 3.2</td><td colspan="5">Requirement Analysis The core objective is to achieve highly accurate semantic segmentation of 3D point clouds, which is a critical capability for advanced systems like autonomous vehicles and complex robotic interaction. The primary challenge being addressed is the inherent unstructured, sparse, and irregular nature of raw 3D point cloud data, which prohibits the direct use of standard CNNs and hinders efficient feature learning. Initial requirements analysis establishes the need for a deep learning architecture that can effectively capture both local geometry and global context despite data sparsity, leading to better performance metrics compared to existing state-of-the-art models. The project culminates in a functional application demo, thus requiring robustness and reasonable inference speed for practical use.

Data Source For rigorous training and unbiased performance evaluation, the project will exclusively utilize widely recognized benchmark datasets for 3D point cloud semantic segmentation (such as S3DIS, ScanNet, PartNet .etc) to assess the model's ability to handle fine-grained part segmentation. These industry-standard datasets provide the necessary scale, diversity, and annotation quality required for developing and validating the novel architecture. Data will be pre-processed to handle variations in density and alignment before being fed into the neural network.

Method The methodology is structured in three phases. First, an extensive investigation of diverse and cutting-edge architectural paradigms will be undertaken to establish strong performance baselines. Second, drawing critical insights from this analysis, a novel deep network architecture will be designed, implemented, and iteratively refined. Finally, the high-performance model will be optimized and packaged for the segmentation application demo.

Implementation The project will culminate in the development of a functional 3D point cloud segmentation application demo. This system will allow a user to upload a raw 3D point cloud file. The system will then process the data using the trained novel deep learning model deployed on the back-end. The immediate output to the user will be a visual representation of the segmented point cloud, where each point is accurately assigned and color-coded according to its predicted semantic class label, demonstrating the direct applicability of the research.

Experiments To prove the project hypothesis—that the novel architecture achieves superior performance—a series of rigorous experiments will be</td></tr>
</table>

	conducted. Comprehensive benchmarking will compare the proposed architecture against established, state-of-the-art baselines across multiple datasets. Performance will be quantitatively measured using a standard set of metrics, including mean Intersection over Union (mIoU), overall classification accuracy, and class-wise precision and recall. Furthermore, specific ablation studies will be performed on our method to validate the role of each component. **Development Tools** The primary programming language for the project will be Python. The deep learning models and all associated architectural components will be implemented using the PyTorch framework, which is selected for its flexibility in custom network design and efficient GPU utilization. Model training and performance testing will primarily utilize an NVIDIA RTX 3090 GPU, ensuring sufficient computational power for handling large 3D datasets and complex model training runs. **Background Materials** The background material consulted for this project includes a rigorous survey of seminal and recent research papers published in major computer vision conferences such as CVPR, ICCV, and ECCV, alongside the latest preprint papers available on Arxiv. This academic literature forms the basis for understanding existing architectures and identifying research gaps. Supplementary materials include specific technical documentation and code repositories related to point cloud processing and deep learning frameworks, as well as the official websites and research reports corresponding to the selected benchmark datasets.
道德规范 **Ethics**	Please confirm by checking the box: ☒ I confirm that I have discussed ethical issues with my supervisor.

Please discuss ethical issues with your supervisor. **Please refer to Project Student Handbook section 4.1 and Project Ethics Guidance**	Summary of ethical issues: **Academic Integrity:** Ensure all foundational research, architectural concepts, code snippets, and ideas from external sources (papers, open-source libraries) are meticulously cited and referenced to strictly avoid plagiarism and fraud. **Data Privacy and Anonymization:** Only utilize publicly available, pre-anonymized benchmark datasets (e.g., S3DIS, ScanNet) that have been ethically vetted. Ensure that no Personally Identifiable Information (PII) or sensitive data is collected, stored, or processed throughout the project's lifecycle. **Algorithmic Bias and Fairness:** Acknowledge the potential for segmentation bias (e.g., lower accuracy on less-represented object classes or specific environments). Ensure that experimental evaluations include comprehensive tests for class-wise performance disparity, incorporating bias detection and mitigation steps within ablation studies. **Transparency and Explainability (Model Opacity):** Clearly document the novel architecture's design principles and feature fusion mechanisms. Aim to improve model transparency by focusing on interpretability metrics to better understand *why* the model makes specific segmentation decisions, especially given the criticality of its potential applications. **Responsible Application and Societal Impact:** The precision and rigorous bias management of Point Cloud Semantic Segmentation are paramount. Errors in segmentation or undetected bias can translate directly into catastrophic failures in high-impact domains such as autonomous driving and advanced robotics. For instance, the misclassification of critical objects (e.g., pedestrians or sudden obstacles) can compromise system safety, leading to operational failures or severe accidents. Consequently, development must strictly adhere to principles that prevent misuse, such as advanced unauthorized surveillance, and focus exclusively on enhancing safety and efficiency within ethically sound applications.
中期目标 **Mid-term target.** **It must be tangible outcomes, E.g. software, hardware or simulation.** **It will be assessed at the mid-term oral.**	**Initial Prototype Implementation:** Implement and fully configure the novel Architecture prototype within the PyTorch framework. **Quantitative Performance Validation:** Train the implemented prototype on an initial split of a primary benchmark dataset and generate verifiable metrics comparing results against some predetermined baselines.

Work Plan (Gantt Chart)

Fill in the sub-tasks and insert a letter X in the cells to show the extent of each task

	Nov 1-15	Nov 16-30	Dec 1-15	Dec 16-31	Jan 1-15	Jan 16-31	Feb 1-15	Feb 16-28	Mar 1-15	Mar 16-31	Apr 1-15	Apr 16-30
Task 1: Conduct a rigorous investigation of diverse architectural paradigms to establish performance baselines												
Survey and categorize existing point cloud segmentation network architectures.	X	X										
Implement baseline models and configure them for benchmark datasets.		X	X									
Conduct initial training and performance validation for selected baseline architectures.		X	X									
Analyze the strengths and weaknesses of different architectural design choices.			X	X								
Task 2: Establish an estimation system to calculate the metrics of an architecture												
Define the core quantitative performance metrics to be calculated.	X	X										
Develop a standardized code pipeline for reproducible training and testing.		X	X									
Design a visualization module for comparative analysis of model outputs and ground truth.		X	X									
Establish protocols for in-depth comparative analysis and ablation studies.			X	X								
Task 3: Using insights from this analysis, design and implement a novel network architecture for enhanced performance.												
Conceptualize and detail the design of the novel architecture.			X	X								
Implement the full novel network architecture in the PyTorch framework.					X	X						
Iteratively train, test, and optimize the novel architecture for target performance gains.						X	X	X				
Conduct comprehensive ablation studies and preliminary interpretability analysis on our method.									X			
Task 4: Build and validate a 3D point cloud segmentation demo to display the applicability clearly.												
Develop the core real-time 3D visualization module for segmented results.										X		
Design and implement a simple, intuitive user interface for the application demo.											X	
Perform comprehensive system testing and validation of the full end-to-end segmentation demo.												X

北京邮电大学 本科毕业设计（论文）初期进度报告

Project Early-term Progress Report

学院 School	International School	**专业 Programme**	**Internet of Things Engineering**		
姓 Family name	Zhao	**名 First Name**	Jian		
BUPT 学号 BUPT number	2022213320	**QM 学号 QM number**	221167928	**班级 Class**	2022215111
论文题目 Project Title	Multi-scale 3D object semantic segmentation based on Deep Learning				

1 Overview

The report is structured as follows: section 2 provides a comprehensive literature review, including a review of existing research and a critical analysis of the field's progress, limitations, and challenges. section 3 outlines the project aims, focusing on addressing key challenges in annotation efficiency, domain generalization, and computational efficiency. section 4 summarizes the progress made so far in the project.

2 Literature Review

Point cloud semantic segmentation is one of the core tasks in 3D scene understanding, and has achieved significant progress in recent years with the development of deep learning techniques. we first review the existing research comprehensively, then provide a critical analysis of the field's progress, limitations, and challenges.

2.1 Review of the Existing Research

This subsection provides a comprehensive overview of existing research in 3D point cloud semantic segmentation, covering datasets, model architectures, pre-training and learning paradigms, open-vocabulary methods, and fine-grained optimization techniques.

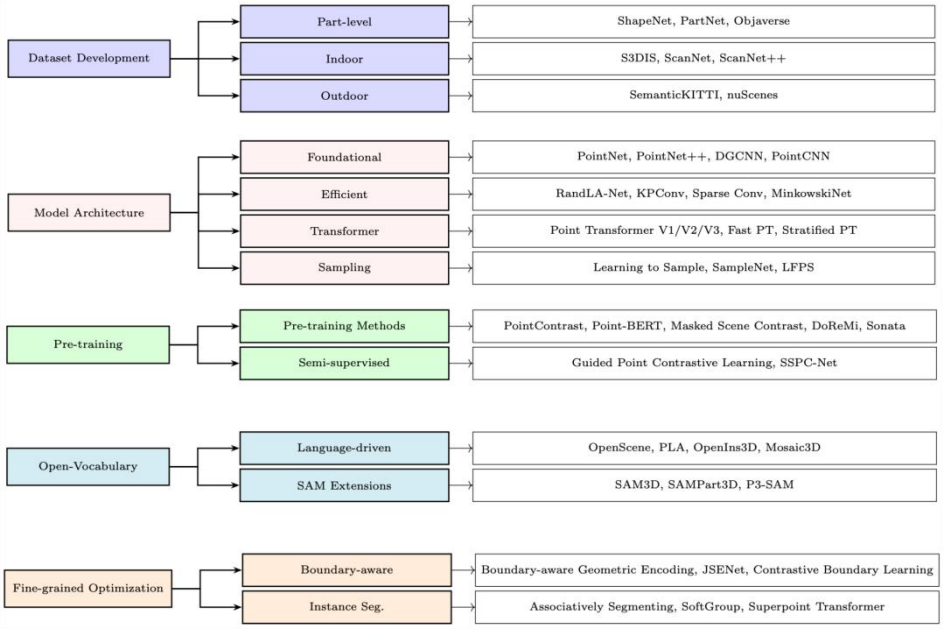


Figure 1: Taxonomy of 3D point cloud understanding methods.

2.1.1 Dataset Development

High-quality datasets serve as a crucial foundation for advancing point cloud segmentation research. Point

cloud segmentation datasets can be broadly categorized into three types: part-level object datasets, indoor scene datasets, and outdoor scene datasets.

Part-level Object Datasets. ShapeNet [4] provides a large collection of 3D shapes with part annotations. PartNet [21] was proposed as a large-scale benchmark for fine-grained and hierarchical part-level 3D object understanding. Objaverse [8] contains a vast collection of annotated 3D objects.

Indoor Scene Datasets. S3DIS [1] was one of the early datasets for 3D semantic parsing of large-scale indoor spaces. ScanNet [7] provides rich annotations for 3D reconstructions of indoor scenes. ScanNet++ [38] offers higher-fidelity 3D indoor scene data. Mosaic3D [17] was introduced as a foundation dataset and model for open-vocabulary 3D segmentation.

Outdoor Scene Datasets. SemanticKITTI [2] is a large-scale dataset for semantic scene understanding of LiDAR sequences in outdoor driving scenarios. nuScenes [3] is a multimodal dataset that includes point cloud semantic annotations for autonomous driving research.

2.1.2 Model Architecture Evolution

The model architectures for point cloud segmentation have undergone significant evolution from foundational networks to Transformer-based approaches.

Foundational Point-based Networks. PointNet [24] first applied deep learning directly on point clouds. PointNet++ [25] introduced hierarchical feature learning. Point-NeXt [26] revisited PointNet++ with improved training strategies. Dynamic Graph CNN (DGCNN) [30] learns local geometric features through dynamic graph convolutional networks. PointCNN [18] implements convolution operations through X-transformation.

Efficient Large-Scale Architectures. RandLA-Net [13] uses random sampling and local feature aggregation for large-scale point clouds. KPConv [27] applies flexible and deformable convolution. Sparse convolution-based methods (Submanifold Sparse Convolutional Networks [11], MinkowskiNet [5]) have shown effectiveness for large-scale scenes. Point-Voxel CNN [19] combines point-based and voxel-based approaches.

Transformer-Based Architectures. Point Transformer [40] successfully applies Transformer architecture to point cloud segmentation. Point Transformer V2 [31] incorporates grouped vector attention mechanisms. Fast Point Transformer [22] addresses computational efficiency. The latest Point Transformer V3 [32] achieves faster speed and stronger performance.

Learnable Sampling Methods. Learning to Sample [9] introduced a learnable sampling approach. SampleNet [16] developed differentiable point cloud sampling. LFPS [12] proposed learned farthest point sampling.

2.1.3 Pre-training and Learning Paradigms

Due to the high cost of point cloud annotation, various learning paradigms have been developed. PointContrast [35] performs unsupervised pre-training through contrastive learning. Point-BERT [39] pre-trains 3D point cloud Transformers using masked point modeling. Multi-dataset point prompt training [33] was proposed for large-scale 3D representation learning. DoReMi [36] is a domain-representation mixture framework for generalizable 3D understanding. Sonata [34] obtains reliable point representations through self-supervised learning.

2.1.4 Open-Vocabulary Segmentation

Open-vocabulary segmentation enables models to segment objects and scenes based on text descriptions. OpenScene [23] enables 3D scene understanding with open vocabularies. Segment Anything Model (SAM) [15] achieved breakthrough progress in 2D image segmentation. SAM3D [37] adapts SAM for 3D scenes. P3-SAM [20] proposes a native 3D part segmentation approach.

2.2 Analysis of Relevant Methods

Based on the comprehensive review above, this subsection provides a critical analysis of the current state of point cloud semantic segmentation, identifying key achievements, limitations, and challenges.

2.2.1 Key Achievements and Progress

The Point Cloud field has witnessed remarkable progress. Model architectures have evolved from point-based networks to Transformer-based approaches, significantly improving accuracy and efficiency. Efficient large-scale architectures enable processing of real-world large-scale scenes. The development of diverse datasets and the emergence of open-vocabulary segmentation have enhanced flexibility and generalizability. Pre-training methods have addressed annotation cost challenges.

2.2.2 Limitations and Challenges

Despite significant progress, several limitations and challenges remain:

1. **Annotation Efficiency.** Creating high-quality annotations for 3D point clouds is still time-consuming and expensive, especially for large-scale outdoor scenes.
2. **Generalization Across Domains.** Models trained on indoor scenes often struggle when applied to outdoor environments, and vice versa. The domain gap between different data sources poses significant challenges for real-world deployment.
3. **Computational Efficiency.** Processing large-scale point clouds in real-time remains challenging, especially for Transformer-based architectures in resource-constrained environments.
4. **Long-tail Distribution.** Most datasets and methods assume relatively balanced class distributions, but real-world scenarios often exhibit significant class imbalance.
5. **Open-Vocabulary Segmentation Limitations.** Current methods often struggle with fine-grained distinctions and may require significant computational resources.

3 Project Aim

Based on the analysis of existing research and identified challenges, we aim to address three critical limitations in point cloud semantic segmentation: annotation efficiency, generalization across domains, and computational efficiency.

Addressing Annotation Efficiency and Domain Generalization. Current approaches typically train models on single-domain datasets, leading to domain-specific overfitting. This limitation not only wastes the valuable annotations available in other domains but also prevents models from learning robust and generalizable representations. Recent work such as multi-dataset point prompt training [33] has made initial progress towards cross-dataset training through domain-specific prompt tuning. However, their method is still limited to indoor scenes, which share relatively similar data distributions. When training on heterogeneous datasets including part-level objects, indoor scenes, and outdoor scenes simultaneously, the data distribution differences become significantly more severe, posing substantial challenges that remain largely unaddressed.

We aim to fuse multiple heterogeneous annotated datasets (e.g., indoor scenes, outdoor scenes, and part-level object datasets) for joint training, addressing the more challenging cross-domain scenarios beyond indoor-only settings. By leveraging complementary information from diverse domains, we expect to improve both annotation efficiency and domain generalization capabilities.

Addressing Computational Efficiency through Adaptive Sampling. When integrating data from different domains for joint training, the segmentation tasks exhibit dramatically different granularity requirements. For instance, when segmenting an indoor scene, the model needs to simultaneously segment coarse-grained structures like walls and fine-grained details like individual keys on a keyboard. This large granularity gap creates a fundamental challenge for efficient processing.

Current methods typically employ uniform grid sampling or uniform learnable sampling strategies (e.g., Learning to Sample [9], SampleNet [16]), which are clearly inadequate for handling such diverse granularity requirements efficiently. These existing learnable sampling methods focus on selecting informative points but do not adaptively control density based on the granularity requirements of different regions.

Unlike structured data that requires fixed grid arrangements (e.g., images for CNNs), point clouds are inherently unstructured and flexible. We exploit this property by treating adaptive sampling as the first layer of the point cloud network architecture, making it differentiable and learnable. The approach

adaptively controls point cloud density: maintaining high density in regions requiring fine-grained segmentation (e.g., object boundaries, detailed parts) while reducing density in coarse-grained regions. This adaptive density control can be naturally supported by fusing fine-grained datasets (part-level) and coarse-grained datasets (indoor/outdoor scenes), enabling efficient segmentation acceleration without sacrificing accuracy in critical regions.

4 Progress Summary

Preliminary Research Directions. A comprehensive literature review has been completed. Based on the literature analysis, preliminary research directions have been proposed. The first direction focuses on heterogeneous dataset fusion, aiming to leverage multiple annotated datasets. The second direction explores differentiable adaptive sampling for point cloud acceleration.

Data Preparation. Essential datasets have been downloaded and preprocessed, including ScanNet [7], S3DIS [1] and so on. These datasets serve as the foundation for validating the proposed heterogeneous dataset fusion approach.

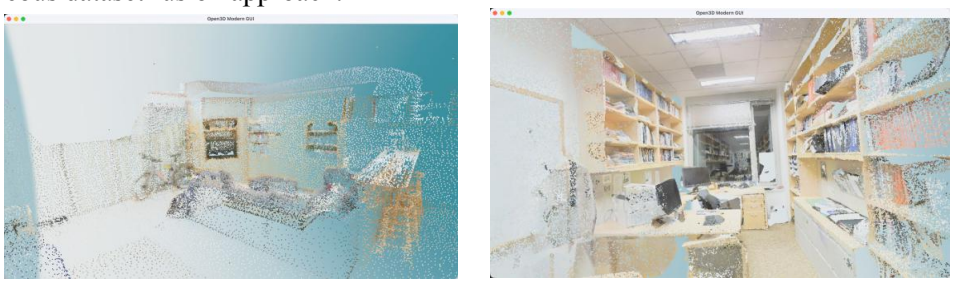


Figure 2: Visualization of the downsampling results of 3D datasets

Framework Development. A comprehensive framework for point cloud semantic segmentation has been established based on the Pointcept library [6]. The framework includes modules for training, inference, and evaluation metric computation, providing a solid foundation for conducting experiments and validating proposed methods.

Initial Experimental Validation. As a preliminary exploration of multi-dataset learning, the work on multi-dataset point prompt training [33] has been initially reproduced on the ScanNet dataset, which helps understand the challenges and opportunities in training models across heterogeneous datasets, providing insights for developing improved methods for heterogeneous dataset fusion.

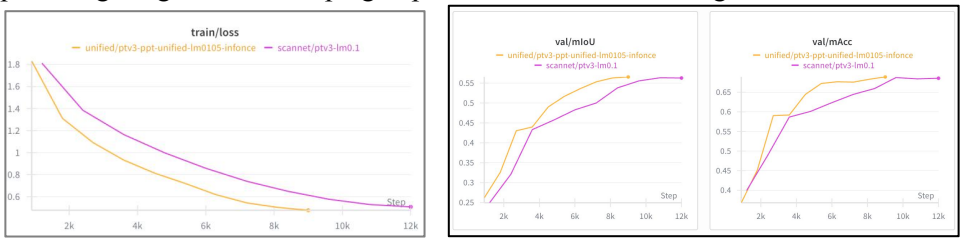


Figure 3: under small dataset configurations, mixed training achieved faster convergence speed.

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是否符合进度？On schedule as per GANTT chart?
YES

下一步 Next steps:
Based on the current progress, the following plan is designed to outline the research steps from the current stage to ablation studies.

Method Development. The first phase focuses on developing the core components of the proposed approach. For heterogeneous dataset fusion, we will design a unified training framework that can effectively handle datasets with different characteristics (part-level objects, indoor scenes, outdoor scenes). This includes developing domain-aware mechanisms to address the significant data distribution differences across heterogeneous datasets. For adaptive sampling, we will implement a differentiable sampling module as the first layer of the network architecture, enabling adaptive density control based on granularity requirements. The sampling module will learn to maintain high density in fine-grained regions while reducing density in coarse-grained regions.

Integration and Training. The second phase involves integrating the heterogeneous dataset fusion and adaptive sampling components into a unified framework. We will develop training strategies that can effectively leverage multiple datasets simultaneously, addressing challenges such as dataset imbalance, different annotation granularities, and varying point cloud densities. The training process will be conducted on the prepared datasets (ScanNet, PartNet, and potentially additional datasets) to validate the effectiveness of the proposed approach.

Baseline Comparisons. The third phase focuses on comprehensive experimental evaluation. We will compare the proposed method against strong baselines, including single-dataset training approaches, existing multi-dataset methods (e.g., PPT), and uniform sampling strategies. Evaluation will be conducted on multiple datasets to assess both in-domain and cross-domain performance, measuring improvements in annotation efficiency, domain generalization, and computational efficiency.

北京邮电大学 本科毕业设计（论文）中期进度报告

Project Mid-term Progress Report

学院 School	International School	**专业 Programme**	**Internet of Things Engineering**		
姓 Family name	Zhao	**名 First Name**	Jian		
BUPT 学号 BUPT number	2022213320	**QM 学号 QM number**	221167928	**班级 Class**	2022215111
论文题目 Project Title	Multi-scale 3D object semantic segmentation based on Deep Learning				

是否完成任务书中所定的中期目标？Targets met (as set in the Specification)?
YES

已完成工作 Finished work:

1 Overview

This report investigates semantic-aware adaptive sampling for 3D point cloud semantic segmentation. The report is structured as follows: 2 provides the preliminary feasibility analysis, including experiment settings, evaluation metrics, and exploratory findings on class imbalance; 3 presents the proposed Sample Net based framework and its training formulation; 4 reports class-wise quantitative results and mechanism analysis; 5 summarizes the current stage and next plan.

2 Preliminary Findings

In the early term report, the project aim is to improve annotation efficiency, domain generalization, and computational efficiency through heterogeneous data usage and adaptive sampling. Following that aim, we conducted experiments to make the feasibility analysis and to determine whether semantic-aware sampling is a reasonable direction.

2.1 Experiment Settings

2.1.1 Datasets

This stage uses ScanNet and S3DIS as the main datasets for exploratory analysis and reproduction experiments (Dai et al. 2017; Armeni et al. 2016). PartNet is kept in the pipeline as a fine-grained dataset for follow-up cross-dataset validation (Mo et al. 2019). Visual examples of the three datasets are shown in 1.

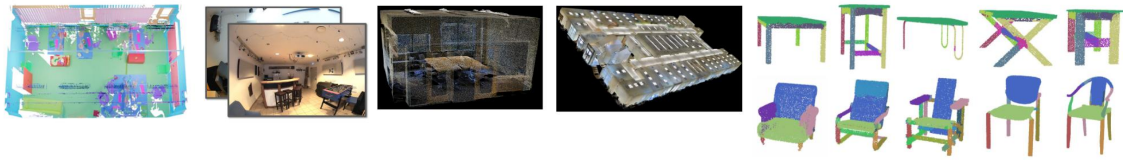


Figure 1 Dataset overview used in this stage: ScanNet(++), S3DIS, and PartNet.

2.1.2 Backbone

The segmentation backbone is Point Transformer V3 (PTV3) (Wu et al. 2024). The architecture workflow used in this report follows the original PTV3 reproduction pipeline, as illustrated in 2.

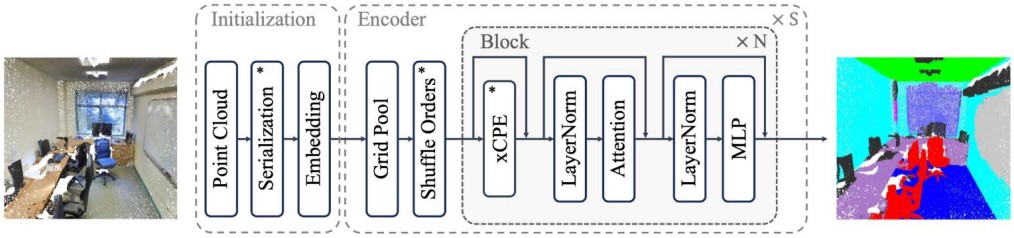


Figure 2 PTV3 backbone framework used for reproduction and comparison.

2.1.3 Sampling Methods

The preliminary setting compares three standard preprocessing sampling strategies:

1. Grid sampling.
2. Random sampling.
3. Farthest point sampling (FPS).

These methods are used as references before introducing semantic-aware adaptive sampling (Dovrat, Lang, and Avidan 2019; Lang, Manor, and Avidan 2020; Heins and Kerschke 2024).

2.1.4 Evaluation Metrics

For semantic segmentation, class-wise IoU and class-wise accuracy are computed first, then averaged over classes:

$$\mathrm{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c + FN_c}, \qquad \mathrm{mAcc} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FN_c}.$$

Here, C is the number of semantic classes, and TP_c, FP_c, FN_c are true positives, false positives, and false negatives for class c.

2.2 Exploratory Design and Findings

We reproduced PTV3 training on ScanNet and S3DIS following the original paper setting,

and then computed class-wise point statistics and class-wise segmentation performance. For a sample point cloud with N points and labels y_i, the point count of class c is:

$$n_c = \sum_{i=1}^{N} \mathbf{1}(y_i = c), \qquad q_c = \frac{n_c}{N}.$$

The statistics show obvious class imbalance in both datasets. On ScanNet, class 0 and class 1 are dominant. On S3DIS, class 0, class 1, and class 2 are dominant. We further observed that this imbalance directly affects class-wise segmentation quality: classes with higher presence tend to obtain better IoU. The correlation analysis gives $r = 0.514$ with $p < 0.05$, and linear fitting explains around one quarter of class-wise IoU variance ($R^2 = 0.264$ for ScanNet and $R^2 = 0.247$ for S3DIS). The corresponding visualizations are shown in 3, with dataset-level average mIoU values 0.7544 (ScanNet) and 0.6256 (S3DIS).

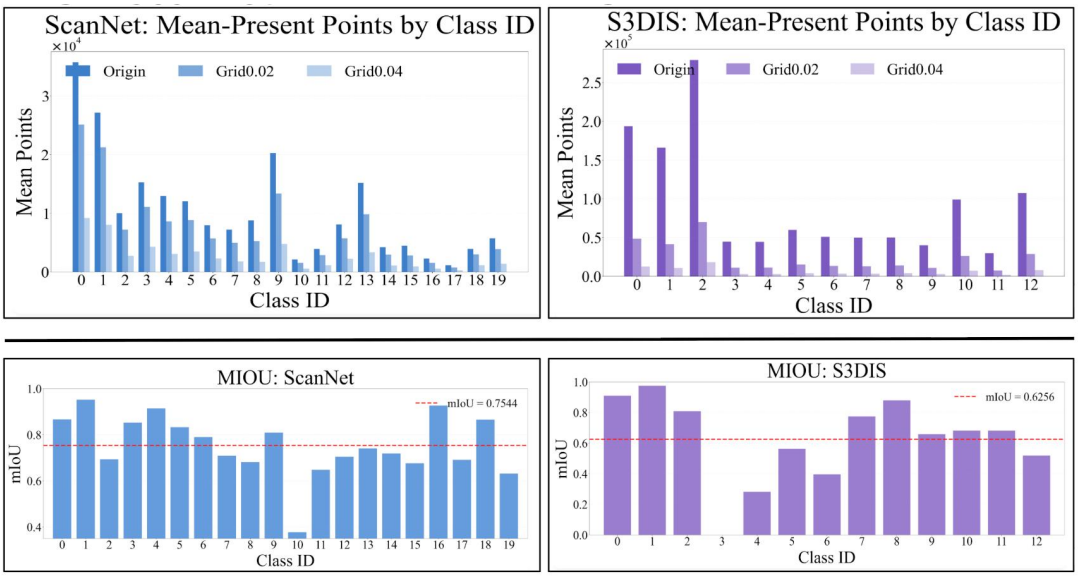


Figure 3 Exploratory statistics on ScanNet and S3DIS: class-wise point distribution under different sampling settings and corresponding class-wise mIoU behavior.

3 Proposed Method

Motivated by the finding that class imbalance is strongly linked to class-wise segmentation performance, we introduce semantic anti-sampling: classes with larger original point counts are assigned lower sampling probability, so that more sampling budget can be shifted to under-represented semantic regions.

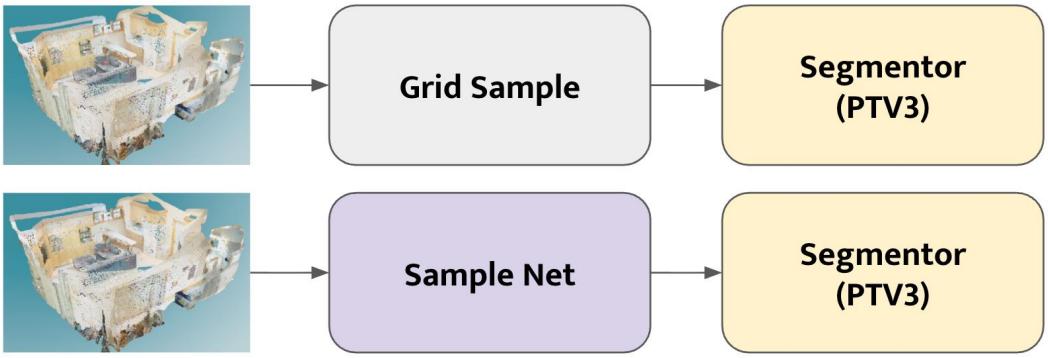


Figure 4 Framework comparison between baseline pipeline and proposed Sample Net based adaptive sampling pipeline.

3.1 Baseline Backbone Training

Given a point cloud $P = \{(x_i, y_i)\}_{i=1}^{N}$, baseline preprocessing applies a fixed sampler $S(\cdot)$ (for example grid sampling) and obtains $\widetilde{P} = \{(\widetilde{x}_j, \widetilde{y}_j)\}_{j=1}^{\widetilde{N}}$. The backbone f_θ outputs class logits $z_j = f_\theta(\widetilde{x}_j)$, and class probabilities are:

$$p_j(c) = \frac{\exp(z_{j,c})}{\sum_{k=1}^{C} \exp(z_{j,k})}.$$

The segmentation objective is the standard per-point supervised loss:

$$\mathcal{L}_{seg} = \frac{1}{\widetilde{N}} \sum_{j=1}^{\widetilde{N}} \ell\left(p_j, \widetilde{y}_j\right),$$

where ℓ is the classification loss used in the PTV3 reproduction setting.

3.2 Sample Net Training

For each sample, statistics are computed independently. Let n_c be the original point count of class c, and $q_c = n_c / N$ be its class proportion. Sample Net g_ϕ predicts a point-wise weight:

$$w_i = g_\phi(x_i), \qquad i = 1, \ldots, N.$$

For point i with label y_i, the supervision target is the semantic proportion of its class:

$$t_i = q_{y_i}.$$

This enforces Sample Net to explicitly encode class-frequency information at point level. The point-wise training objective is written as:

$$\mathcal{C}_{weight} = \frac{1}{N}\sum_{i=1}^{N} \ell_w(w_i, t_i),$$

where ℓ_w is the weight regression loss.

To implement semantic anti-sampling, point sampling probability is inversely related to semantic proportion. At class level, budget allocation satisfies:

$$\sum_{c=1}^{C} \min(n_c, \tau) = M, \qquad m_c = \min(n_c, \tau),$$

where M is the total sampling budget, τ is the threshold satisfying the budget constraint, and m_c is the allocated sampled count for class c. The corresponding class-level keep ratio is:

$$r_c = \frac{m_c}{n_c}.$$

Thus larger n_c leads to smaller r_c, which directly realizes anti-sampling for dominant classes.

The backbone and Sample Net are optimized under:

$$\mathcal{L} = \mathcal{L}_{seg} + \lambda \mathcal{L}_{weight},$$

where λ balances segmentation supervision and weight supervision.

4 Experiment Results

Following the metrics defined in 2, we report class-wise performance on ScanNet subset-0.1. The full comparison is shown in Table 1.

Method	Metric	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	Average
Point Count	$\times 10^3$	35.8	27.2	10.0	15.3	13.0	12.0	8.0	7.2	8.8	20.3	2.2	3.9	8.1	15.2	4.2	4.5	2.3	1.1	4.0	5.8	10.4
Grid Sampling	mIoU	77.1	93.6	43.6	**71.2**	80.6	61.5	57.7	44.2	45.9	**68.1**	26.0	53.3	47.3	**66.5**	40.5	55.3	83.4	48.9	81.3	28.6	58.7
	mAcc	**92.8**	97.4	62.0	**77.4**	91.0	74.9	73.0	57.9	58.7	77.1	32.1	60.0	66.7	**74.4**	44.8	62.2	92.7	56.5	85.5	34.2	68.6
Orcale Adaptive	mIoU	**80.1**$^{\uparrow 3.0}$	**94.8**$^{\uparrow 1.2}$	**48.9**$^{\uparrow 5.2}$	65.8$^{\downarrow 5.4}$	**82.4**$^{\uparrow 1.8}$	**67.6**$^{\uparrow 6.1}$	64.6$^{\uparrow 6.9}$	**63.7**$^{\uparrow 19.4}$	**63.8**$^{\uparrow 17.9}$	64.5$^{\downarrow 3.6}$	**51.2**$^{\uparrow 25.2}$	**61.9**$^{\uparrow 8.6}$	**56.1**$^{\uparrow 8.8}$	60.4$^{\downarrow 6.1}$	**47.0**$^{\uparrow 6.5}$	**72.6**$^{\uparrow 17.3}$	**89.3**$^{\uparrow 5.9}$	55.8$^{\uparrow 6.9}$	84.8$^{\uparrow 3.5}$	**38.9**$^{\uparrow 10.3}$	**65.7**$^{\uparrow 7.0}$
	mAcc	90.3$^{\downarrow 2.5}$	**97.9**$^{\uparrow 0.5}$	**64.4**$^{\uparrow 2.4}$	73.6$^{\downarrow 3.8}$	**92.5**$^{\uparrow 1.6}$	**79.2**$^{\uparrow 4.2}$	80.0$^{\uparrow 7.0}$	**81.5**$^{\uparrow 23.7}$	**80.1**$^{\uparrow 21.4}$	**82.8**$^{\uparrow 5.6}$	59.8$^{\uparrow 27.7}$	70.7$^{\uparrow 10.7}$	**75.4**$^{\uparrow 8.7}$	72.0$^{\downarrow 2.4}$	**56.9**$^{\uparrow 12.1}$	**86.2**$^{\uparrow 24.0}$	**98.6**$^{\uparrow 5.9}$	66.0$^{\uparrow 9.5}$	92.2$^{\uparrow 6.7}$	**49.0**$^{\uparrow 14.9}$	**77.5**$^{\uparrow 8.9}$
Sample Net	mIoU	63.8$^{\downarrow 13.3}$	93.5$^{\downarrow 0.1}$	41.6$^{\downarrow 2.0}$	61.8$^{\downarrow 9.4}$	78.2$^{\downarrow 2.4}$	54.4$^{\uparrow 7.1}$	**64.8**$^{\uparrow 7.1}$	58.1$^{\uparrow 13.8}$	51.9$^{\uparrow 6.0}$	56.2$^{\downarrow 11.9}$	43.5$^{\uparrow 17.5}$	55.6$^{\uparrow 2.3}$	47.1$^{\downarrow 0.2}$	49.9$^{\downarrow 16.6}$	35.0$^{\downarrow 5.5}$	59.4$^{\uparrow 4.1}$	87.4$^{\uparrow 3.9}$	**56.3**$^{\uparrow 7.3}$	**87.5**$^{\uparrow 6.2}$	37.2$^{\uparrow 8.6}$	59.2$^{\uparrow 0.4}$
	mAcc	76.6$^{\downarrow 16.2}$	97.2$^{\downarrow 0.2}$	57.2$^{\downarrow 4.8}$	66.9$^{\downarrow 10.5}$	89.6$^{\downarrow 1.3}$	69.1$^{\downarrow 5.8}$	**84.0**$^{\uparrow 10.9}$	79.1$^{\uparrow 21.3}$	76.3$^{\uparrow 17.6}$	65.6$^{\downarrow 11.5}$	**76.4**$^{\uparrow 44.3}$	**71.1**$^{\uparrow 11.1}$	68.1$^{\uparrow 1.3}$	58.2$^{\downarrow 16.1}$	53.5$^{\uparrow 8.7}$	78.5$^{\uparrow 16.3}$	98.1$^{\uparrow 5.4}$	**67.3**$^{\uparrow 10.8}$	**94.8**$^{\uparrow 9.3}$	45.1$^{\uparrow 10.9}$	73.6$^{\uparrow 5.1}$

Table 1 ScanNet class-wise comparison on subset-0.1. Values in mIoU/mAcc are percentages (%). Point counts are reported in $\times 10^3$. Bold marks the best value in each column. For Orcale Adaptive and Sample Net, superscript arrows indicate the change relative to Grid-Sampling.

Class-to-semantic mapping is: 0-wall, 1-floor, 2-cabinet, 3-bed, 4-chair, 5-sofa, 6-table, 7-door, 8-window, 9-bookshelf, 10-picture, 11-counter, 12-desk, 13-curtain, 14-refridgerator, 15-shower curtain, 16-toilet, 17-sink, 18-bathtub, and 19-otherfurniture.

Mechanism analysis shows a clear trade-off pattern. Compared with Grid-Sampling, Sample Net improves several lower-count semantics, for example 7 (door), 10 (picture), 17 (sink), 18 (bathtub), and 19 (otherfurniture). At the same time, some high-frequency or context-sensitive semantics drop, such as 0 (wall), 5 (sofa), 9 (bookshelf), and 13 (curtain). A plausible explanation is that semantic anti-sampling reallocates limited points from dominant classes to under-represented classes, improving long-tail categories but reducing coverage for some large-structure or context-dependent categories. This is consistent with the gap that still exists between Sample Net and Oracle Adaptive on average metrics.

5 Conclusion

This mid-term study confirms that class imbalance is a key factor affecting class-wise segmentation quality, and supports the feasibility of semantic-aware adaptive sampling. The current Sample Net already improves average accuracy over Grid-Sampling, while still trailing Oracle Adaptive. The next step is to narrow this gap, extend validation to S3DIS and PartNet, and build a real-time visualization demo.

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尚需完成的任务 Work to do:

Finalize Sample Net training design
The semantic anti-sampling idea is in place, but the training objective still needs to be fully specified and stabilized. The next step is to finalize the point-wise supervision, budget control, and integration with segmentation loss into a reproducible training pipeline.

Extend validation to more datasets
Current quantitative results are mainly on a ScanNet subset. We still need systematic experiments on S3DIS and PartNet to verify whether the method generalizes across scenes and semantic granularities.

Complete analysis and demo deliverables
Beyond the main result table, we should add stronger ablations, category-group analysis, and qualitative error cases. In parallel, we need to build the real-time visualization demo to show practical behavior, not only offline metrics.

存在问题 Problems:

Gap to Oracle Adaptive remains significant
Although Sample Net improves over Grid-Sampling on average, it still trails Oracle Adaptive on both mIoU and mAcc. This indicates that the learned policy has not yet reached the upper-bound allocation quality.

Class-level gains are uneven
Some long-tail classes improve, while several frequent or structure-critical classes degrade. This shows a trade-off in the current redistribution strategy between helping underrepresented classes and preserving dominant-class stability.

Higher training and tuning complexity
Adding Sample Net increases optimization complexity and hyperparameter sensitivity. Without a more standardized tuning protocol, iteration speed slows down and result interpretation becomes less consistent.

拟采取的办法 Solutions:

Refine the objective and regularization
Improve the Sample Net loss design and coupling with segmentation loss, and add clearer budget-aware regularization. This should make sampling decisions more task-driven and reduce unstable optimization behavior.

Adopt stratified semantic allocation
Use class-group-aware or dynamic-threshold sampling instead of one uniform anti-sampling rule. This can preserve key high-frequency structures while still improving low-frequency categories.

Use a staged experiment protocol

Run fast screening on a controlled small setup first, then scale to full training and multi-dataset validation. In each stage, pair quantitative metrics with visualization analysis so that each design change has a clear performance explanation.

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Chinese Keywords
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北京邮电大学 本科毕业设计（论文）教师指导记录表

Project Supervision Log

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论文题目 Project Title	Multi-scale 3D object semantic segmentation based on Deep Learning				

Date: 27-08-2025
Supervision type: online meeting
Summary: discussed the project background and possible research directions

Date: 07-11-2025
Supervision type: online meeting
Summary: discussed submission requirements and process management

Date: 10-11-2025
Supervision type: messages
Summary: received written feedback on the project specification

Date: 06-01-2026
Supervision type: messages
Summary: received feedback on the early-term report

Date: 14-01-2026
Supervision type: online meeting
Summary: discussed experiment challenges and provided instructions on the early-term report

Date: 02-03-2026
Supervision type: online meeting
Summary: discussed experiment results and provided instructions on the mid-term presentation

Date: 04-03-2026
Supervision type: messages
Summary: received feedback on the mid-term presentations

Additional appendices (as needed)

Training Hyperparameters

Table 9: Default hyperparameters for the three-stage learned-sampler pipeline.

Parameter	Stage 1	Stage 2	Stage 3
Target budget	Dataset-specific matched budget: Grid004 for ScanNet/S3DIS, and the PartNet fixed-subset reference for PartNet.	Oracle marginal probabilities derived from the corresponding dataset-specific water-filling reference.	Materialised learned-sampler indices calibrated to the same dataset-specific budget.
Selector / subset	Label-aware oracle water-filling sampler.	No fixed subset during training; the sampler predicts point-wise keep probabilities.	Fixed sampler-index subset for each scene.
Trained model	Standard PTV3 downstream segmentor.	Lightweight PTV3-style sampler.	Standard PTV3 downstream segmentor.
Input / supervision	Oracle-selected points with ground-truth semantic labels.	Point coordinates, colour, and normal features when available; supervised by oracle marginal probabilities.	Sampler-selected points with ground-truth semantic labels.
Objective	Cross-entropy + Lovász-Softmax.	Probability MSE against oracle marginal keep probabilities.	Cross-entropy + Lovász-Softmax.
Initialisation	Standard segmentor initialisation.	Standard sampler initialisation.	Stage 1 oracle-selected segmentor checkpoint when applicable.
Optimizer	AdamW, weight decay 0.05.	AdamW, weight decay 0.05.	AdamW, weight decay 0.05.
Scheduler	OneCycleLR, cosine annealing, warm-up ratio 0.05.	OneCycleLR, cosine annealing, warm-up ratio 0.05.	OneCycleLR, cosine annealing, warm-up ratio 0.05.
Learning rate	6×10^{-3} for the segmentor and 6×10^{-4} for PTV3 block parameters.	3×10^{-3}.	Same as Stage 1.
Training length	800–1600 epochs, depending on the dataset.	2000 epochs.	800–1600 epochs, depending on the dataset.
Evaluation interval	Every 10 epochs.	Every 5 epochs.	Every 10 epochs.
Effective batch size	12–24, depending on GPU memory.	12–24, depending on GPU memory.	12–24, depending on GPU memory.
Precision	Mixed precision with FP16.	Mixed precision with FP16.	Mixed precision with FP16.

Related Work

We summarize the research threads most relevant to the present project, namely benchmark development, backbone design for point cloud semantic segmentation, data-efficient learning, open-vocabulary understanding, fine-grained optimisation, class imbalance mitigation, and point sampling.

Datasets and benchmark development. Progress in point cloud semantic segmentation has been driven not only by better models but also by the gradual expansion of benchmark datasets. At the object level, ShapeNet provided a large repository of 3D models [34], and PartNet later introduced fine-grained hierarchical part annotations for detailed object understanding [29]. More

recently, large-scale object collections such as Objaverse further increased the breadth of available 3D assets [37]. At the scene level, S3DIS and ScanNet became the standard indoor semantic segmentation benchmarks [12, 1], while ScanNet++ improved scene fidelity and annotation quality for modern 3D perception pipelines [10]. For outdoor applications, SemanticKITTI and nuScenes established large-scale benchmarks built around autonomous driving and LiDAR sensing [6, 7]. More recently, Mosaic3D began to connect dataset construction with foundation-model style open-vocabulary segmentation [38]. Taken together, these datasets reveal a persistent fact that is central to this report: class frequency is highly uneven across scenes, object scales, and sensing domains, so imbalance is not a corner case but a structural property of 3D data.

Segmentation backbones. Early progress in point cloud semantic segmentation was driven by point-based networks that operate directly on unordered point sets. PointNet established a permutation-invariant framework based on shared multilayer perceptrons and symmetric aggregation [4], while PointNet++ introduced hierarchical neighbourhood grouping to better capture local geometric structure [5]. Subsequent work improved local relation modelling through dynamic graph construction and convolution-like operators, such as DGCNN, PointCNN, PointConv, and KPConv [14, 39, 13, 32]. Hybrid and efficient backbones such as PVCNN, Minkowski-style sparse convolutions, RandLA-Net, and PointNeXt focused on scaling to larger scenes while preserving accuracy [40, 30, 24, 41]. More recently, transformer-based models, including Point Transformer, Stratified Transformer, Fast Point Transformer, Point Transformer V2, and Point Transformer V3, have shown stronger ability to combine local geometry with wider contextual reasoning [15, 42, 43, 33, 11]. Diffusion-based segmentation has also started to appear, suggesting another possible direction for richer conditional modelling [44]. These developments suggest that the main segmentation backbones are already mature enough that performance bottlenecks increasingly come from data quality, supervision efficiency, and preprocessing choices rather than from a lack of expressive network capacity.

Data-efficient learning and cross-domain generalisation. Because dense annotation for benchmarks such as ScanNet and S3DIS is expensive and time-consuming [1, 12], a large body of work has explored how to reduce reliance on fully labelled data. Early representation-learning approaches such as Info3D and PointContrast used contrastive objectives to learn transferable geometric features from unlabeled data [45, 46]. Later methods, including Point-BERT, Point-MAE, and Masked Scene Contrast, showed that masked modelling and scalable scene-level pre-training can also produce useful representations for downstream segmentation [47, 23, 48]. Recent work has further expanded toward reliable self-supervision and broader data coverage, for example through Sonata, multi-dataset point prompt training, and domain-mixture learning frameworks such as DoReMi [49, 50, 51]. In semi-supervised settings, Guided Point Contrastive Learning and SSPC-Net exploit both labelled and unlabelled scenes to reduce annotation cost [22, 52]. These methods improve representation quality before or during fine-tuning and help

the backbone extract stronger geometric cues from limited supervision. However, they do not directly resolve the imbalance introduced when some semantic categories occupy far more points than others inside a scene.

Open-vocabulary and promptable 3D segmentation. Another active research direction aims to remove the fixed label-set limitation of classical semantic segmentation. OpenScene and PLA align 3D scene features with language representations so that segmentation can respond to text queries rather than only a pre-defined class list [53, 54]. OpenIns3D extends this idea toward open-vocabulary instance segmentation [55], while diffusion-assisted open-vocabulary segmentation further explores using generative priors to bridge visual and linguistic concepts [56]. In parallel, Segment Anything [57] inspired a family of promptable 3D methods, including SAM3D, SAMPart3D, and P3-SAM [58, 59, 60]. Mosaic3D also fits into this broader trend by linking foundation-scale data curation with open-vocabulary 3D segmentation [38]. Although these methods are not designed specifically for long-tailed class imbalance, they highlight a broader lesson: future 3D segmentation systems will likely need more flexible supervision and more adaptive data selection mechanisms than standard closed-set pipelines.

Fine-grained optimisation and scene understanding. Beyond global architecture design, several lines of work improve segmentation quality by targeting specific failure modes. Boundary-aware methods explicitly enhance class transitions and edge regions, where point cloud predictions are often unstable because of sparsity and mixed local geometry. Representative examples include boundary-aware geometric encoding, JSENet, and contrastive boundary learning [61, 62, 63]. Other work focuses on richer scene decomposition and instance-level reasoning. Associatively Segmenting Instances and Semantics, SoftGroup, and Superpoint Transformer each improve the separation of neighbouring objects or superpoints in cluttered scenes [64, 65, 66]. Geometry-disentangled representation learning further aims to separate complementary geometric cues for better understanding of complex structures [67]. These methods show that even strong backbones still struggle in precisely the regions where semantic evidence is sparse, boundaries are thin, or object categories are small. This observation is highly relevant to the present project, because such regions are also the ones most vulnerable to aggressive or geometry-only sampling.

Class imbalance mitigation. A standard response to semantic imbalance is to modify the optimisation objective after the input point set has already been formed. Weighted cross-entropy increases the contribution of rare classes, focal loss emphasises hard examples instead of already easy predictions [16], and Lovász-Softmax aligns optimisation more closely with IoU-oriented evaluation [17]. In point cloud segmentation, these losses are often combined with augmentation and large-scene training strategies [30, 35]. More recent 3D-specific methods further target long-tailed learning through imbalance-aware objectives and prototype-based guidance [19, 20]. These approaches are valuable because they reduce bias toward dominant categories such as walls and floors. Nevertheless, they mostly intervene at the loss or feature level. If the

upstream sampler has already discarded too many points from minority classes, later reweighting can only compensate partially because the missing geometric evidence is no longer present.

Point sampling as a research gap. Sampling is unavoidable in large-scale point cloud segmentation because raw scenes usually contain far more points than a backbone can process directly. Classical strategies such as grid sampling, random sampling, and farthest point sampling are designed around geometric coverage or efficiency rather than semantic balance [5, 24]. As a result, large structural regions tend to preserve a disproportionate share of the point budget, while small or rare semantic classes may become even less visible after preprocessing. Learnable sampling methods have begun to move beyond hand-crafted heuristics. Learning to Sample and differentiable point-cloud samplers use differentiable modules to produce task-oriented point subsets [26, 27], and learned farthest point sampling further relaxes the rigid geometric rule of FPS [28]. However, the main goal of these methods is usually to preserve geometric fidelity or general task performance with fewer points. Comparatively little work asks whether the sampling stage itself can be used to counter semantic imbalance before feature extraction begins.

Taken together, the literature indicates a clear opportunity. Benchmark construction, backbone design, self-supervised pre-training, promptable segmentation, and imbalance-aware losses have each improved point cloud segmentation substantially, but the interaction between *sampling* and *semantic class imbalance* remains underexplored. This gap directly motivates the present project: instead of only correcting imbalance after downsampling, we investigate whether a lightweight learnable sampler can reallocate the point budget away from dominant classes and toward under-represented regions at the preprocessing stage.

Risk and Environmental Impact Assessment

This section summarises the main risks and potential environmental impacts associated with the project and outlines the corresponding mitigation actions. For each event, L denotes the likelihood of occurrence, C denotes the seriousness of consequence, and the overall risk score is calculated as $R = L \times C$. Following the project handbook, the risks are grouped into five levels: no risk (0), low risk (1–3), moderate risk (4–6), significant risk (8–12), and high risk (15–25).

Risks to Successful Project Completion

The main completion risk is interruption to the experimental pipeline. Training and evaluating deep learning models for point cloud semantic segmentation requires long GPU runs, repeated ablation studies, and reliable storage for checkpoints and logs. Compute or storage failure, together with insufficient time for re-running key experiments, is assessed as $L = 3$, $C = 4$, and $R = 12$, which is significant. The mitigation is to keep the codebase under version control, save checkpoints regularly, back up essential results, and prioritise the core experiments before optional extensions.

Potential Harm to People and Animals

This project does not involve human participants, animal testing, or deployment in a live robotic system, so the direct project-time risk to people and animals is low. The main concern is indirect downstream harm if an insufficiently validated model were applied in safety-critical settings such as autonomous driving or robotics. Misclassification of pedestrians, obstacles, or under-represented classes could contribute to unsafe downstream decisions. This risk is assessed as $L = 2$, $C = 4$, and $R = 8$, which is significant. The mitigation is to present the model strictly as a research prototype, report class-wise performance and failure cases, and avoid overstating deployment readiness.

Data privacy risk is low because the project uses only public benchmark datasets that are already anonymised and does not collect personally identifiable information. This is assessed as $L = 1$, $C = 2$, and $R = 2$. A related issue is model bias and opacity: some classes or environments may be less well represented, and the model is not fully interpretable by default. This is assessed as $L = 2$, $C = 3$, and $R = 6$, which is moderate. The mitigation is to include class-wise evaluation, discuss performance disparity explicitly, and document the model design clearly enough that readers can understand its limitations.

Environmental Impact

The project does not involve hazardous materials, physical manufacturing, or laboratory waste. Its main environmental impact comes from electricity use associated with GPU training, hyper-parameter tuning, and repeated evaluation. This is assessed as $L = 3$, $C = 2$, and $R = 6$, which is moderate. Large checkpoints and duplicated outputs also create a smaller storage footprint risk, assessed as $L = 2$, $C = 2$, and $R = 4$. The mitigation is to start with small pilot runs, reuse checkpoints where possible, terminate clearly unpromising runs early, and keep only essential outputs.

Potential Financial Loss

The financial exposure of the project is limited, but repeated long-running experiments still consume shared compute resources and can create avoidable cost if hardware fails or results must be regenerated. This is assessed as $L = 2$, $C = 3$, and $R = 6$, which is moderate. The mitigation is to preserve important intermediate results, maintain regular backups, use institutional resources responsibly, and keep a clear and prioritised experiment schedule so that the most informative runs are completed first.

Overall Assessment

Overall, the project does not present a high immediate safety risk, but it does carry meaningful risks to successful completion and to the responsible downstream use of the model. These risks remain manageable provided that experiments are carefully planned, only public anonymised datasets are used, class-wise bias is reported transparently, and the system is presented as a research study rather than as a deployment-ready product.